

C.1 ISTORIA MAȘINILOR ARTIFICIALE ȘI EVOLUȚIA SECURITĂȚII CIBERNETICE

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PRINCIPALELE PĂRȚI ALE PREZENTĂRII

C.2 Istoria mașinilor artificiale și evoluția securității cibernetice:

- C.1.1 ISTORIA MAȘINILOR ARTIFICIALE DE UZ GENERAL
- C.1.2 EVOLUȚIA ALPICAȚIILOR MALWARE
- C.1.3 EVOLUȚIA SOLUȚIILOR DE SECURITATE
- C.1.4 SISTEMELE DE OPERARE ȘI STRATUL DE PORTABILITATE

C.1.1

ISTORIA MAȘINILOR ARTIFICIALE DE UZ GENERAL



Prima placă grafică(I) – semnal digital (1745)



Jacques de Vaucanson



Basile Bouchon

- Prima mașină de țesut cu memorie binară pe carduri!

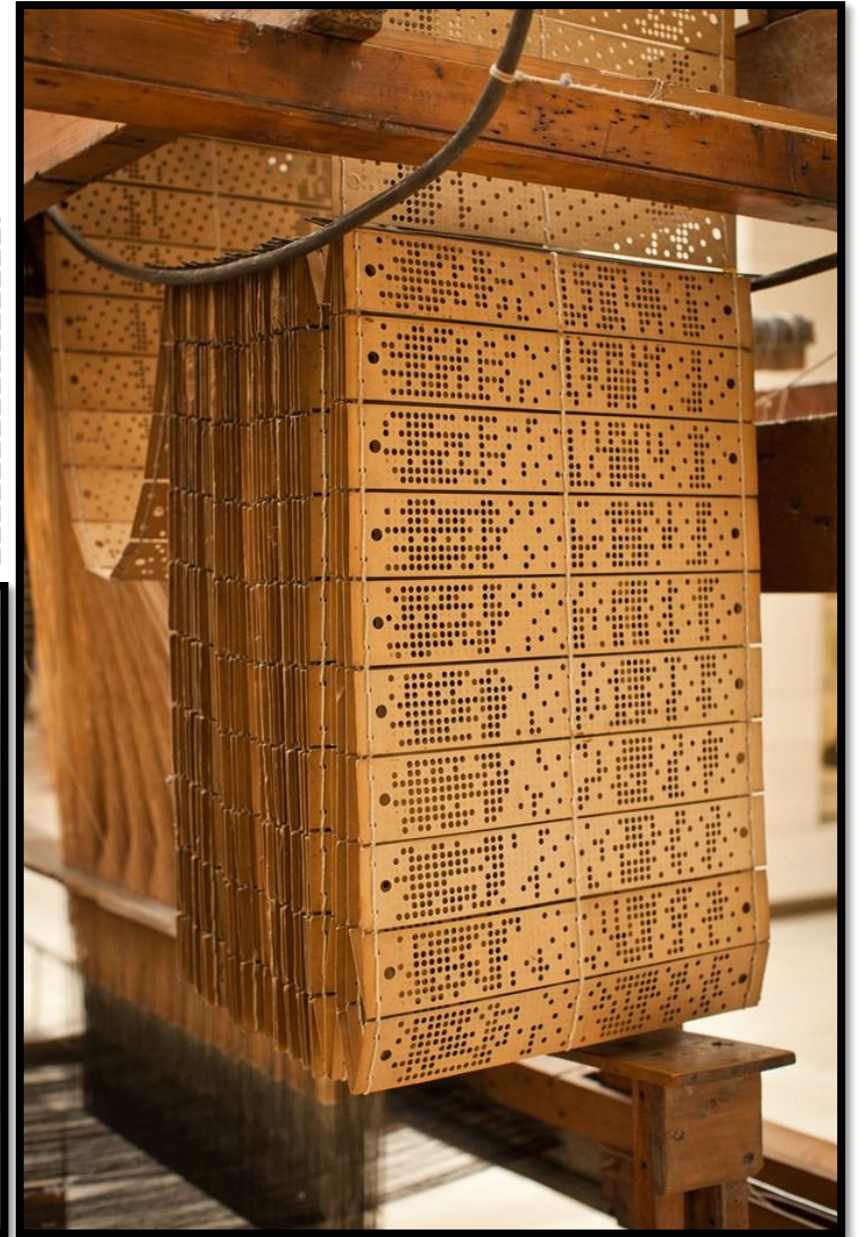
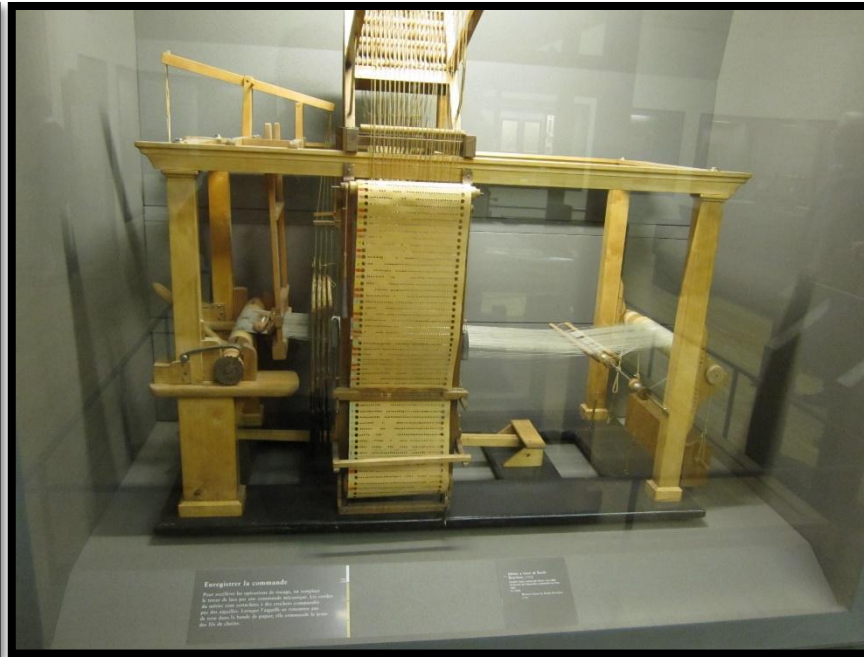
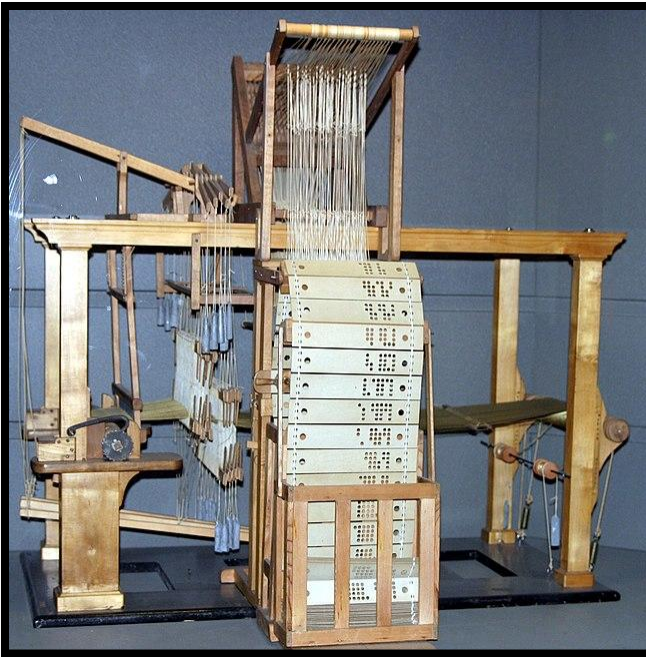
Prima imagine digitala din lume !



Imaginea este doar un exemplu fictiv!

Prima placă grafică (II) - 1745

- Gaură - 1
- Lipsă - 0





Roboți mecanici cu memorie (roboți - 1774)

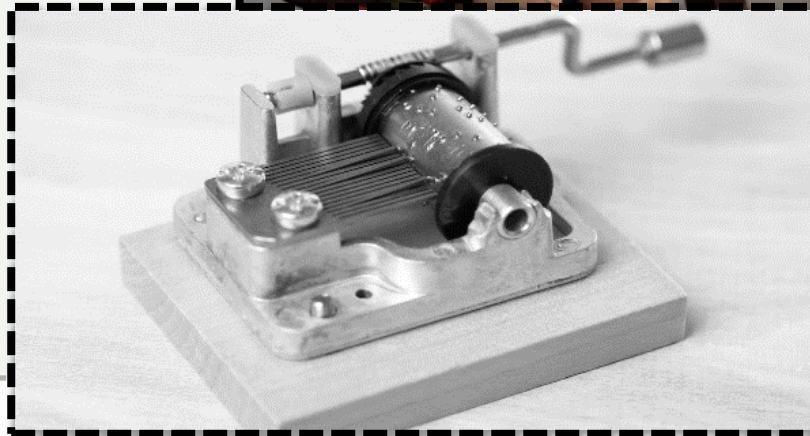
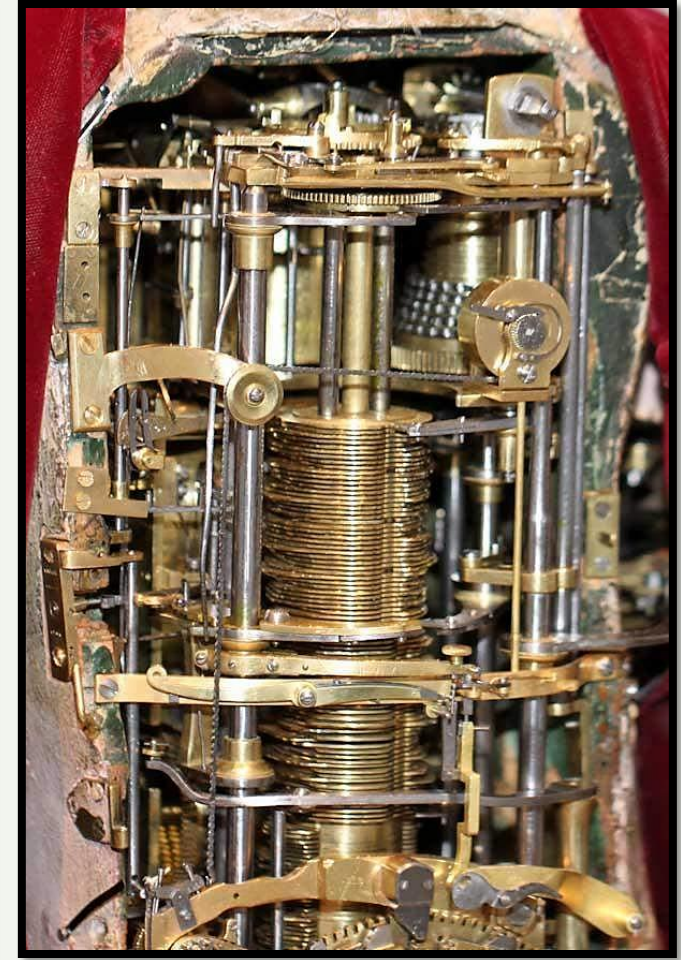
The writer
(1774)



The writer (1774)

Roboți mecanici cu memorie (roboți - 1774)

- Semnal analog.
- 30-40 de caractere.
- memorie – ax cu discuri ale căror proeminențe împing un braț mecanic.
- Parțial asemănător cu cilindrul unei cutii muzicale.



Semnal digital

Calculatoare mecanice (uz general)

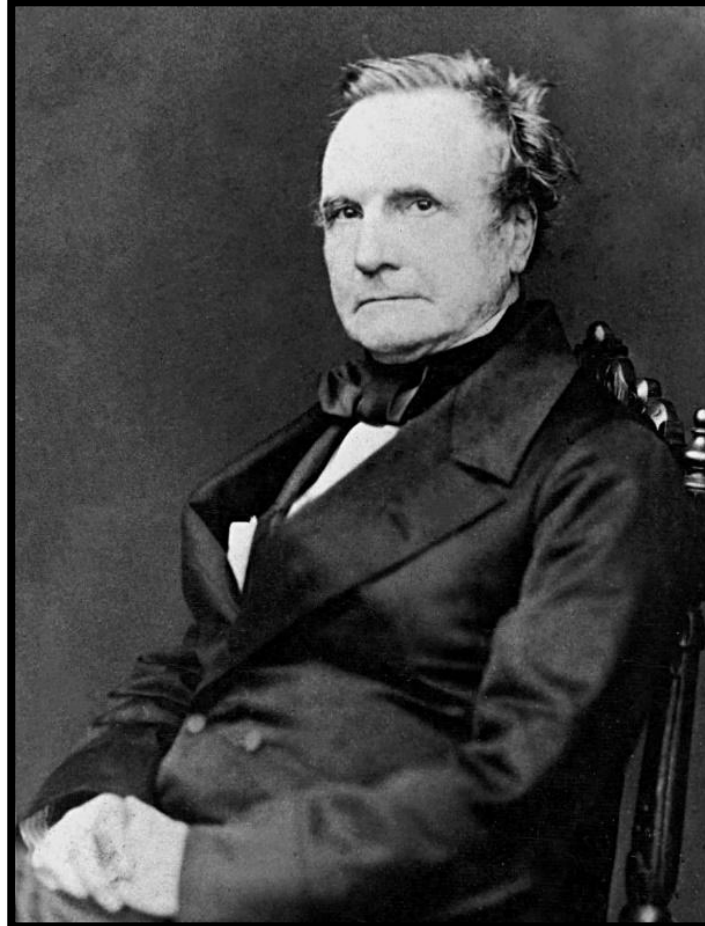
Sfarsitul revolutiei
industriale

Charles Babbage;
"Motorul Analitic"
1830

Dupa
106 de ani

Konrad Zuse; Z1
1936

Calculatorul theoretic "Motorul Analitic" - 1830



Charles Babbage (hardware)
Ada Byron (programmer)

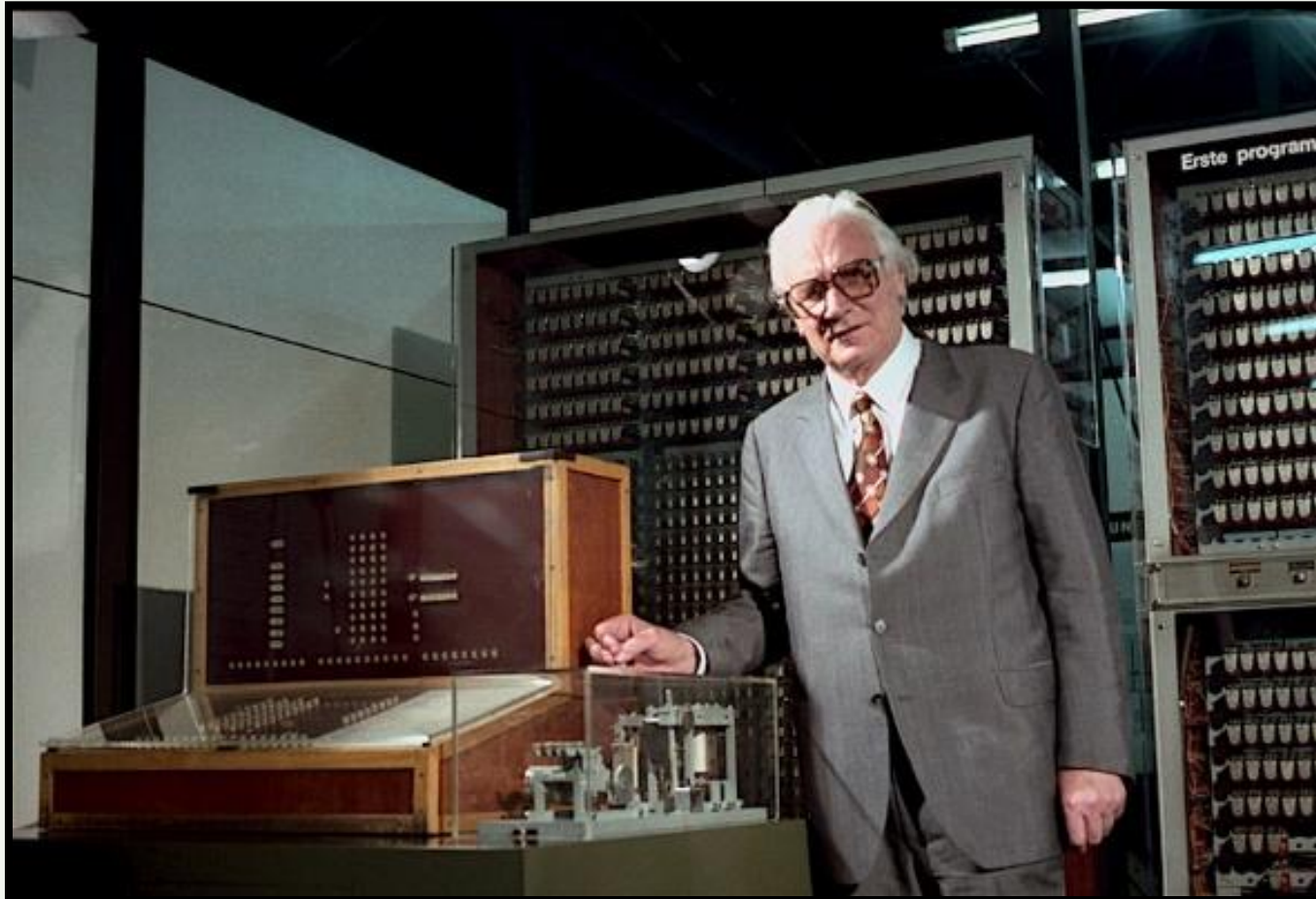
Calculatorul Z1 - 1936



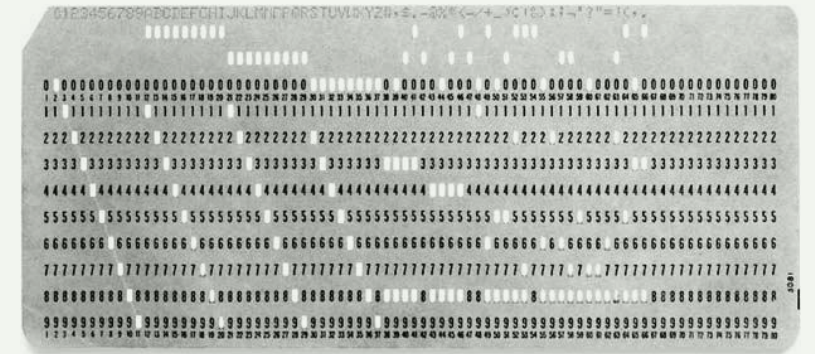
Konrad Zuse

Calculatoare electro-mecanice

Calculatorul Z3 -1941; 5 ani mai târziu...



Konrad Zuse



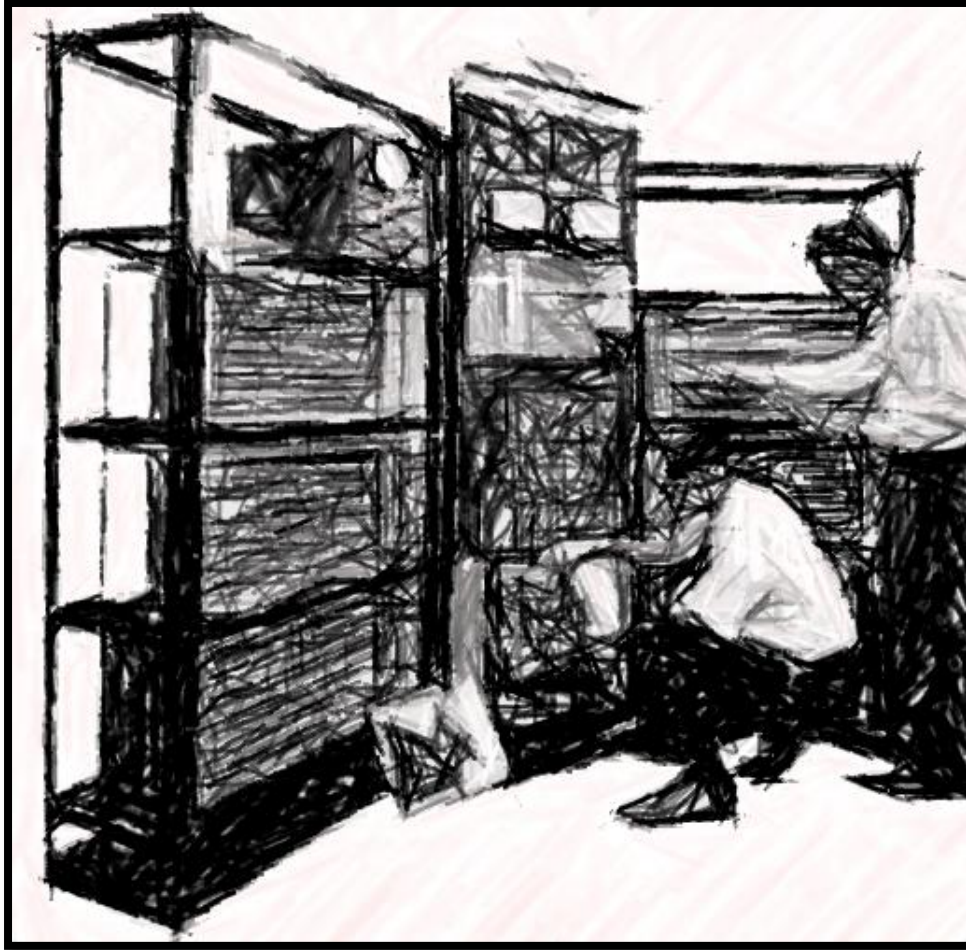
Releu electro-mecanic

Calcolatore elettronice

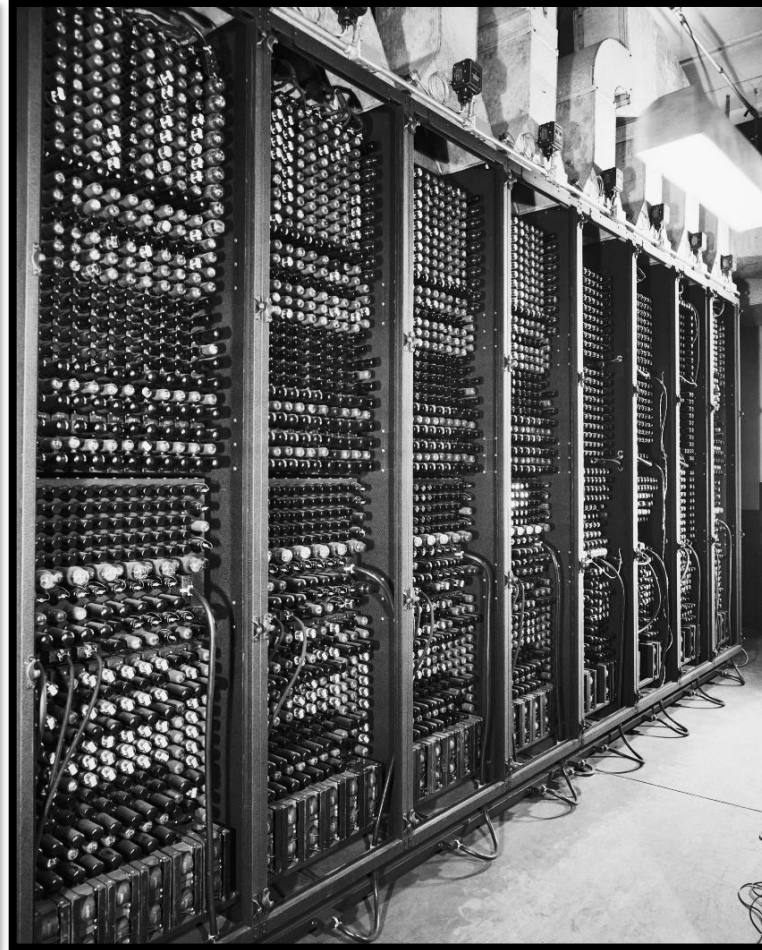
TRADIC – 1950 (pe 800 tranzistoare)



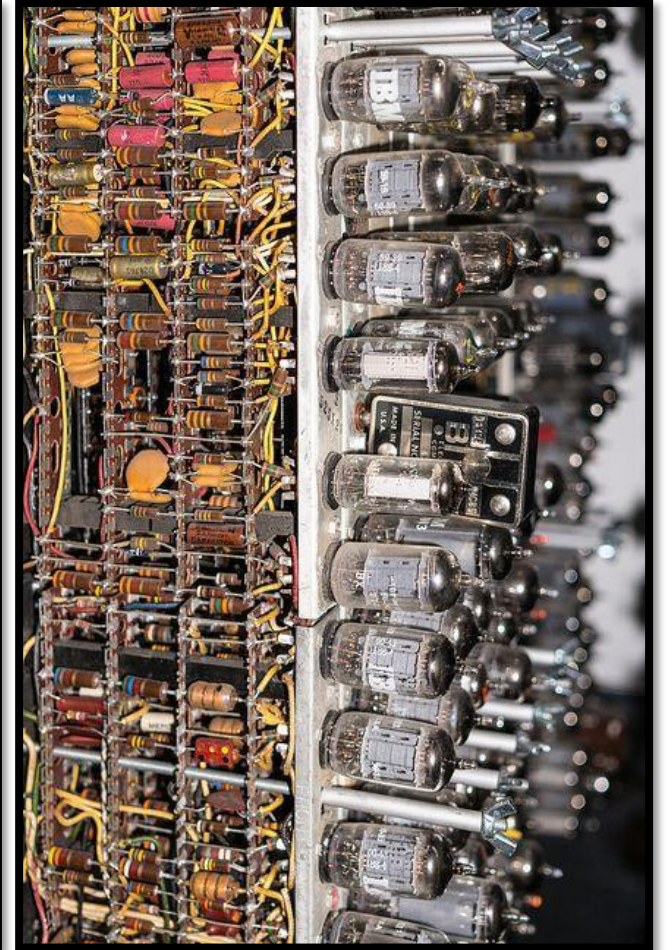
ENIAC – 1945 (pe 17k lampi)



(Transistorized Airborne Digital Computer)



(Electronic Numerical Integrator and Computer)



Calculatoare electronice (piese discrete)



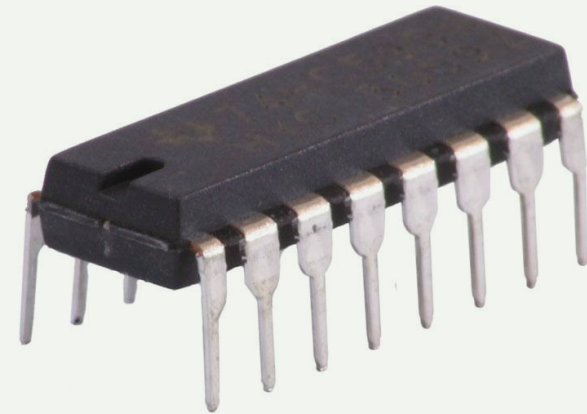
Releul
1835



Tuburi vidate
1907



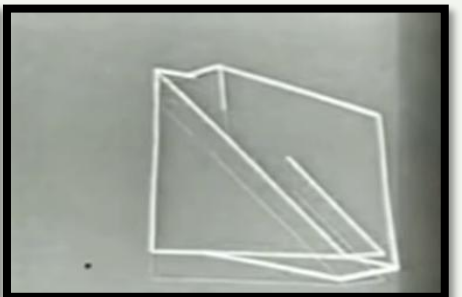
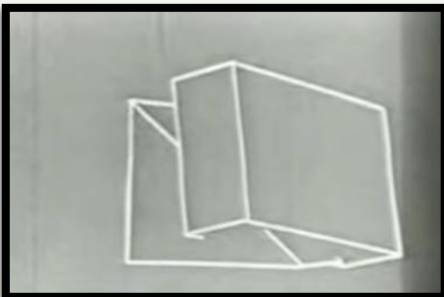
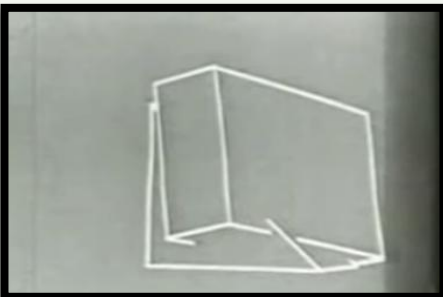
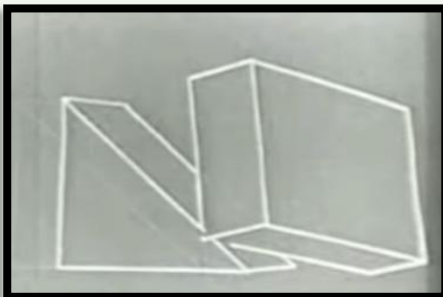
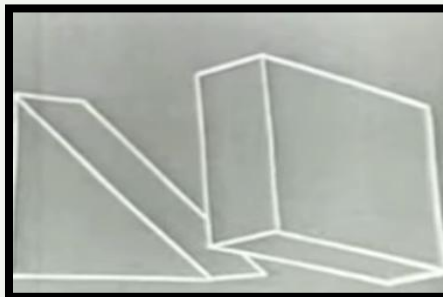
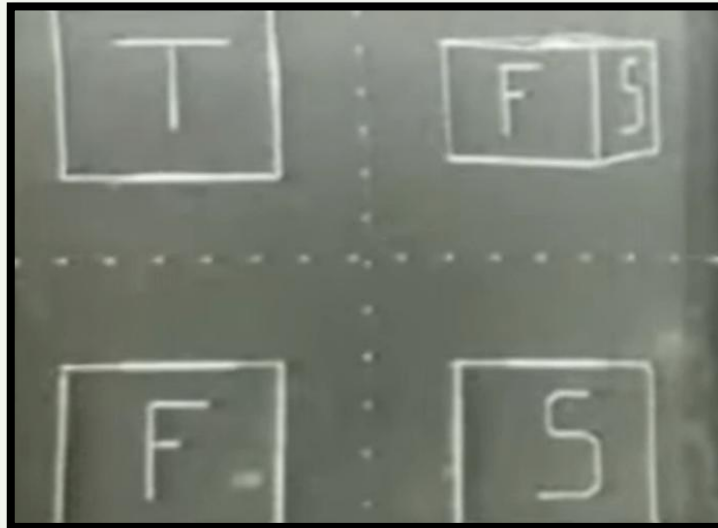
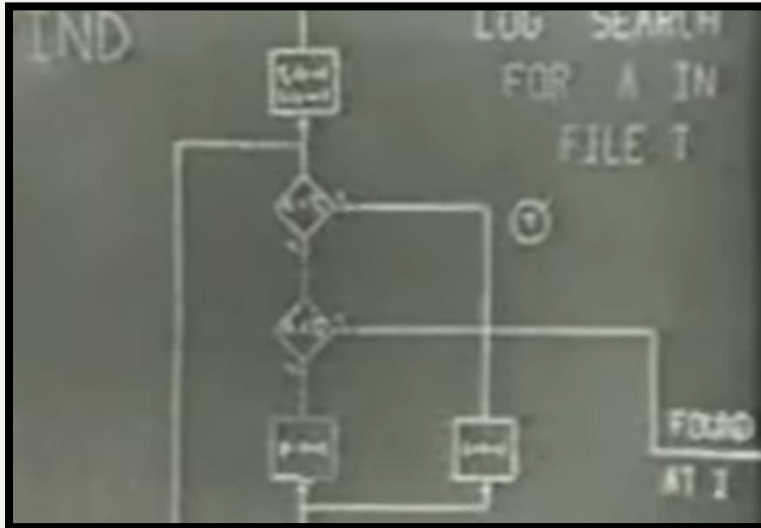
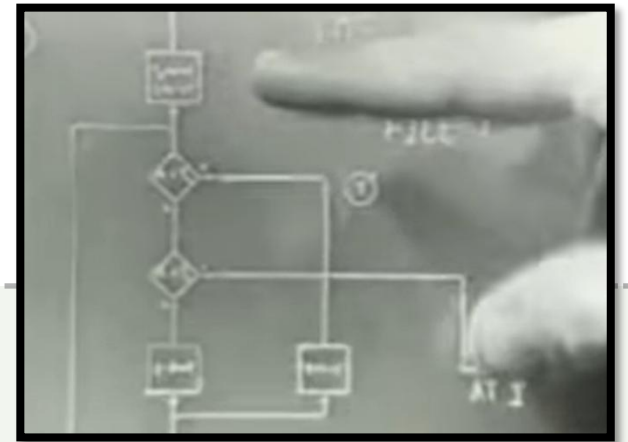
Tranzistori
1947



Circuit Integrat
1963

Prima interfata grafica (GUI)

- 1962 (diagrame, obiecte 3D ...)



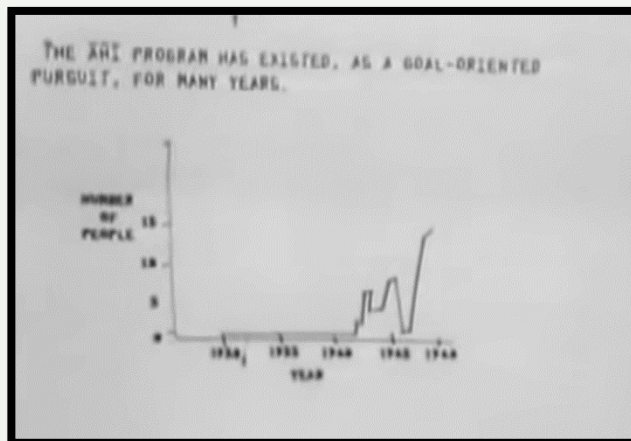
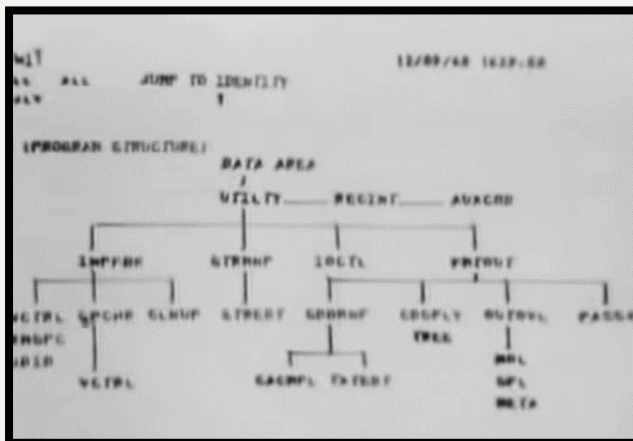
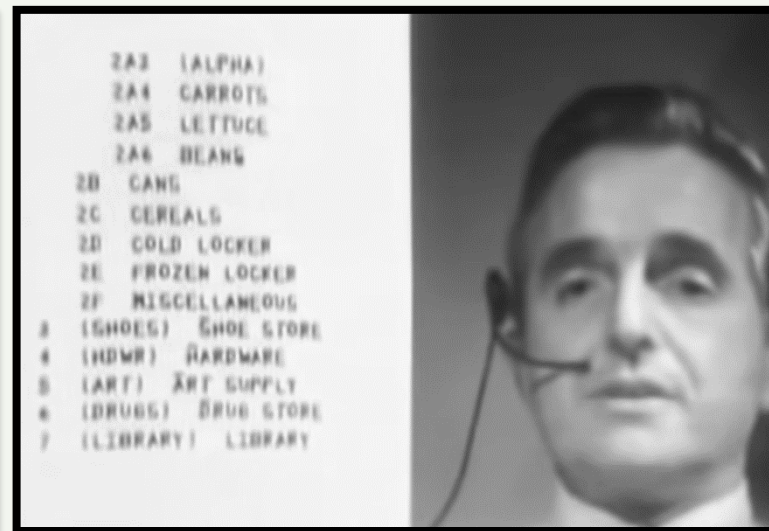
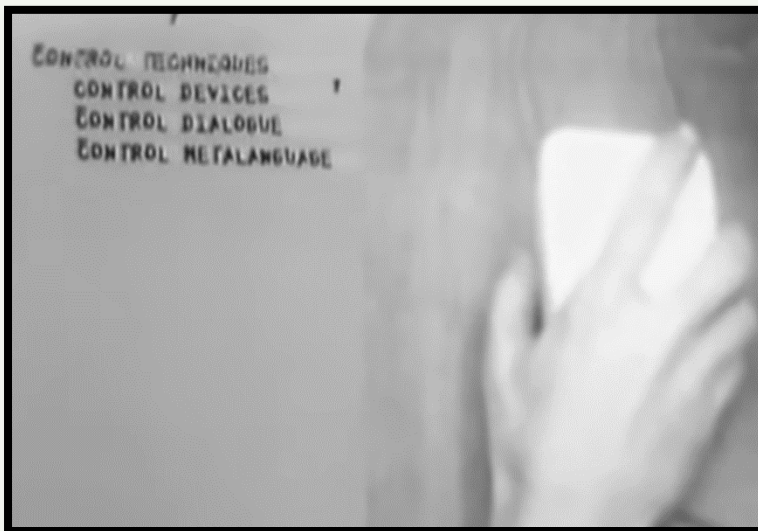
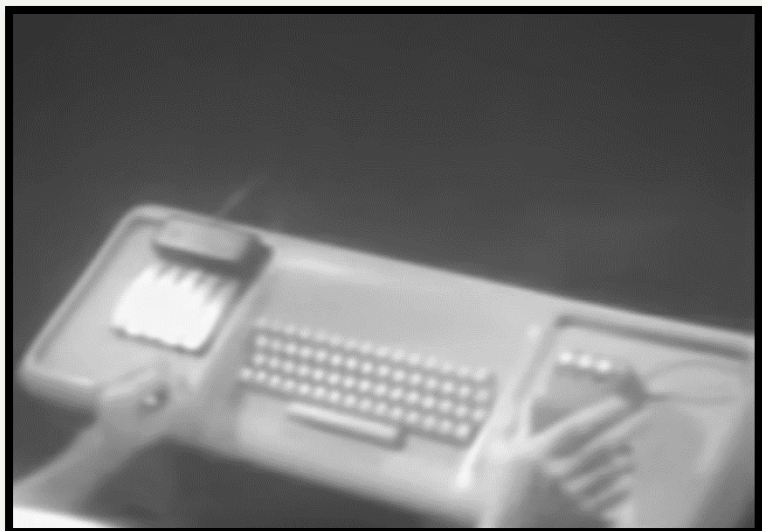
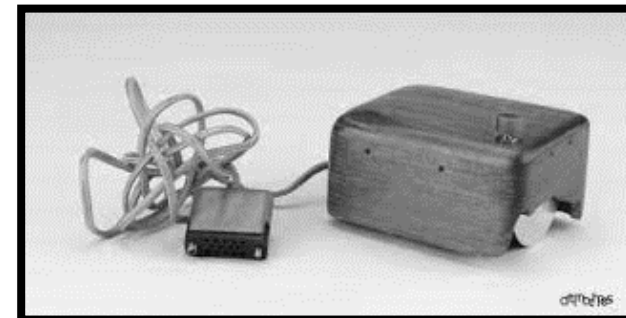
940 SDS

1968



- Douglas Engelbart a inclus conceptul de hipertext, care este un element de bază al HTML și al World Wide Web.
- Augmentarea (îmbunătățirea în latină) a interacțiunii om-mașină.

Douglas Engelbart - Decembrie 9, 1968:
Mother of all demos



ASCII

Line
Feed
(LF:10)



Carriage
Return
(CR:13)

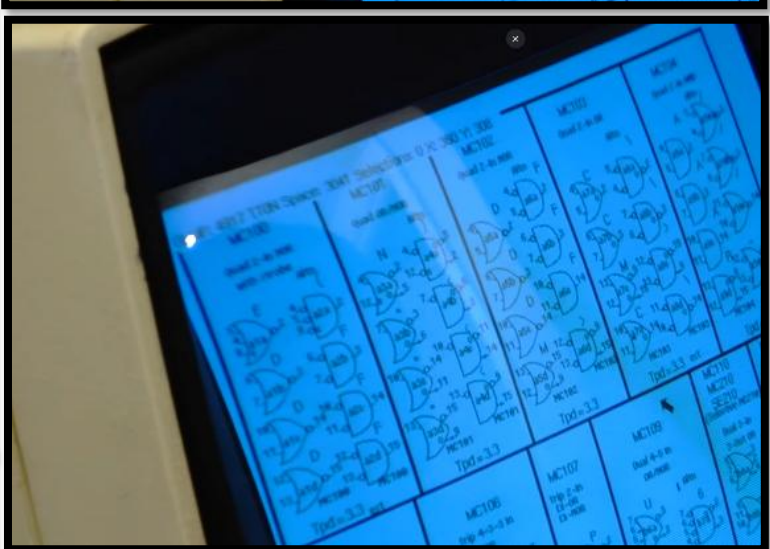
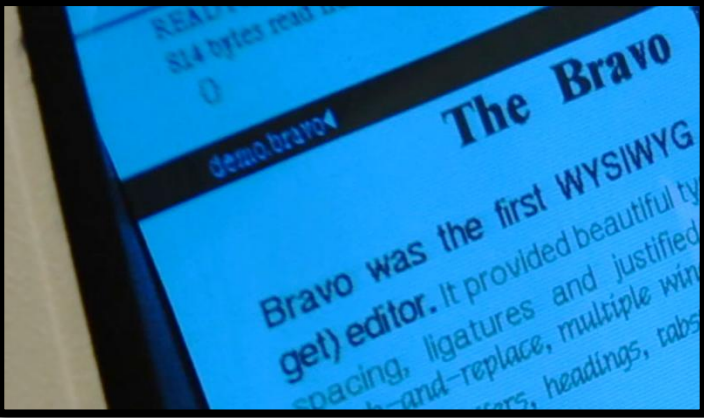
*American Standard
Code for Information
Interchange* – (128 caractere; versiunea extinsa 255 caractere)

Poziția tastelor pe tastaturile moderne?



Ce reprezintă ASCII?

Xerox Alto
1973



Software modern

Alto Operating System (AOS)

Sursa tutoror desktop-urilor !

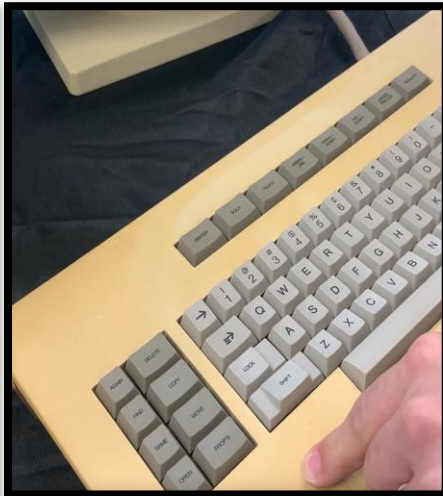
The left photograph shows a white evidence marker (1) and a ruler next to a white card. The card contains a list of items and a table with columns for 'Item', 'Date', and 'Time'.

The right photograph shows a white evidence marker (1) and a ruler next to a white card. The card contains a list of items and a table with columns for 'Item', 'Date', and 'Time'.



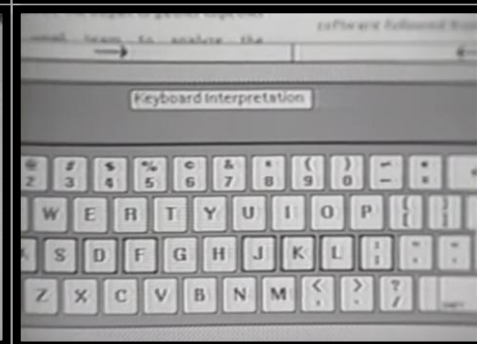
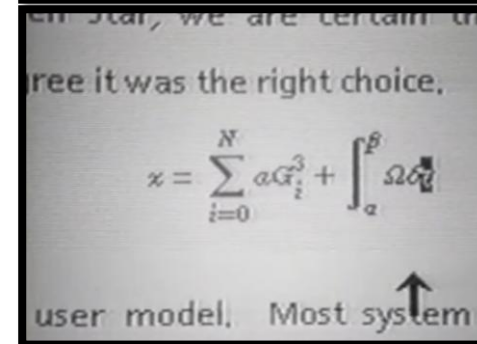
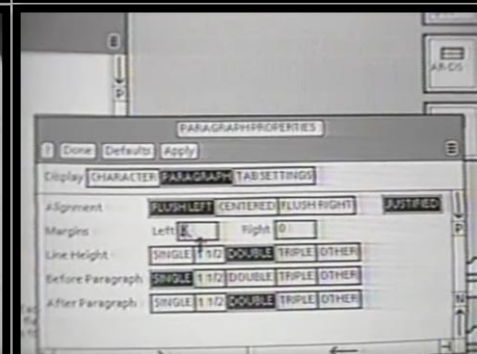
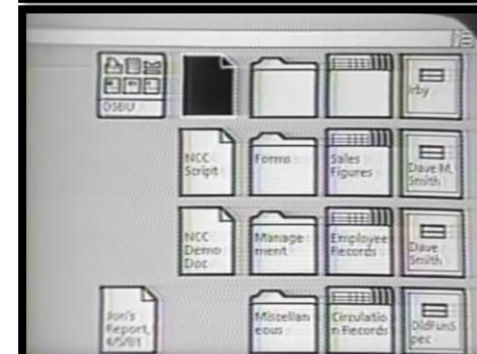
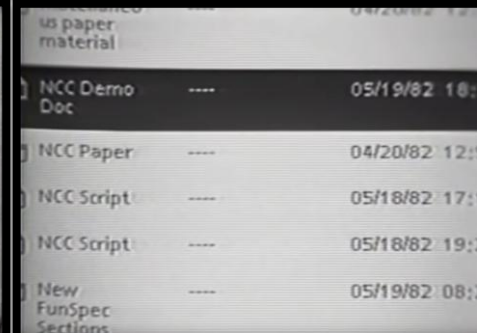
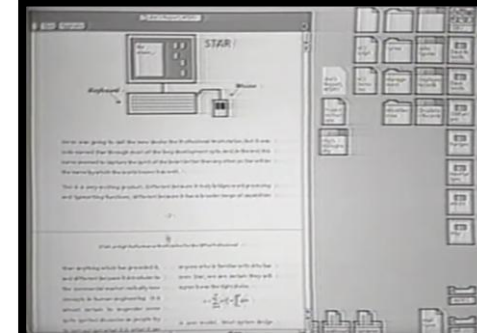
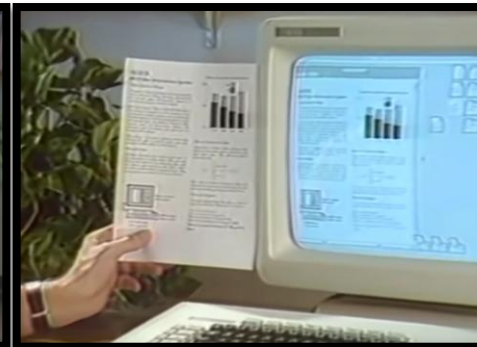
Xerox Star 8010 Computer

1981

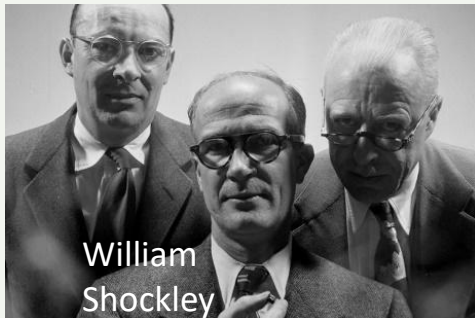
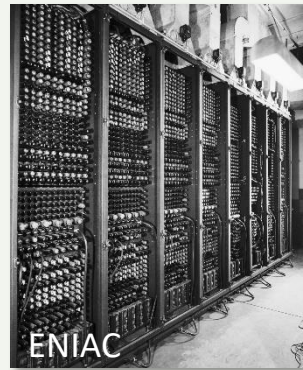
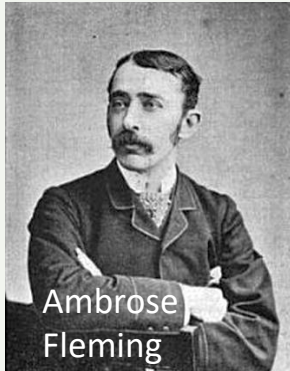
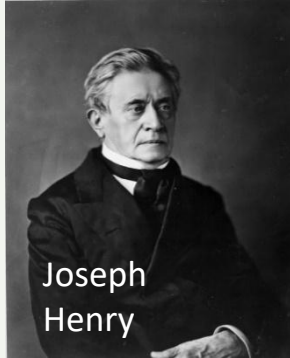


Xerox Star OS
16-bit
8Mhz

Ce observam?



Recapitulare ! De la releu la tranzistor !



- Releu
- Lămpi
- Tranzistoare
- CI (Circuite Integrate)

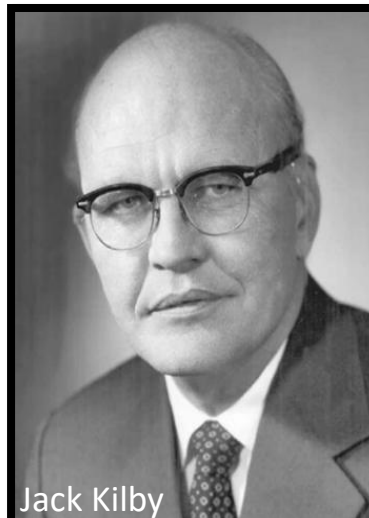
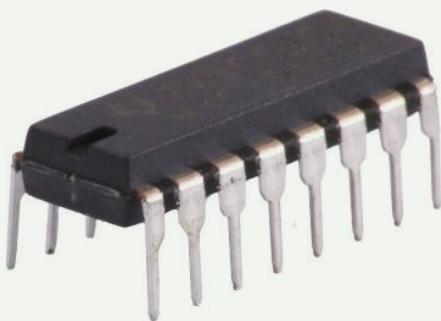
Recapitulare ! Era semiconductorilor !

Descoperirea efectului de tranzistor

Compactarea in Circuite Integrate (CI)

Primele calculatoare desktop

William Shockley



Jack Kilby



foto-litografie
IC modern

Robert Noyce



C.1.2

EVOLUȚIA APLICAȚIILOR MALWARE



Cand apar primele aplicatii malware?

Phone

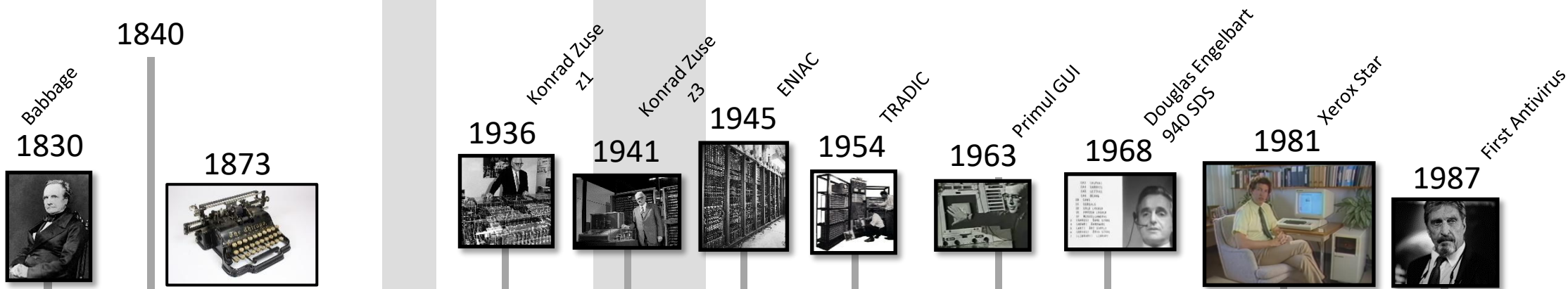
Calculatoare personale (PC)

(1953) Mainframes (calculator central de mare putere)

Rev. Industrială

Calculatoare mecanice si electro-mecanice

Calculatoare electronice



Astăzi



1835
Codul Morse (1837)
si telegraful (1844)



1907



1947

American Standard
Code for Information
Interchange – 128 chr



1963



1972

Xerox Alto

99% din toate fenomenele fizice au fost descoperite

WWI
(1914-1918)

WWII
(1939-1944)

Apariția treptată a
aplicațiilor malware
din 1971

1972



Xerox Alto

1981



Xerox Star

*File Virus;
MS-DOS;

Cascade
1987

.exe
(resident)

*Backdoor;
VAX/VMS
OS;

Vienna
1987

.com
(non-resident)

WANK
1989

Virusul Elk Cloner: Un poem era afisat la fiecare 50 de infectari.

1971
Creeper

*Worm;
TENEX OS;
ARPANET

1975
Animal

*Trojan;
UNIVAC 1100
OS

Mainframe

1982
Elk Cloner

*Boot Virus;
Apple II OS

1986
Brain

*Resident
Boot Virus;
MS-DOS

1987
Stoned

Boot Virus;
MS-DOS

1988
Morris

Worm;
UNIX;
*(resident
RAM)

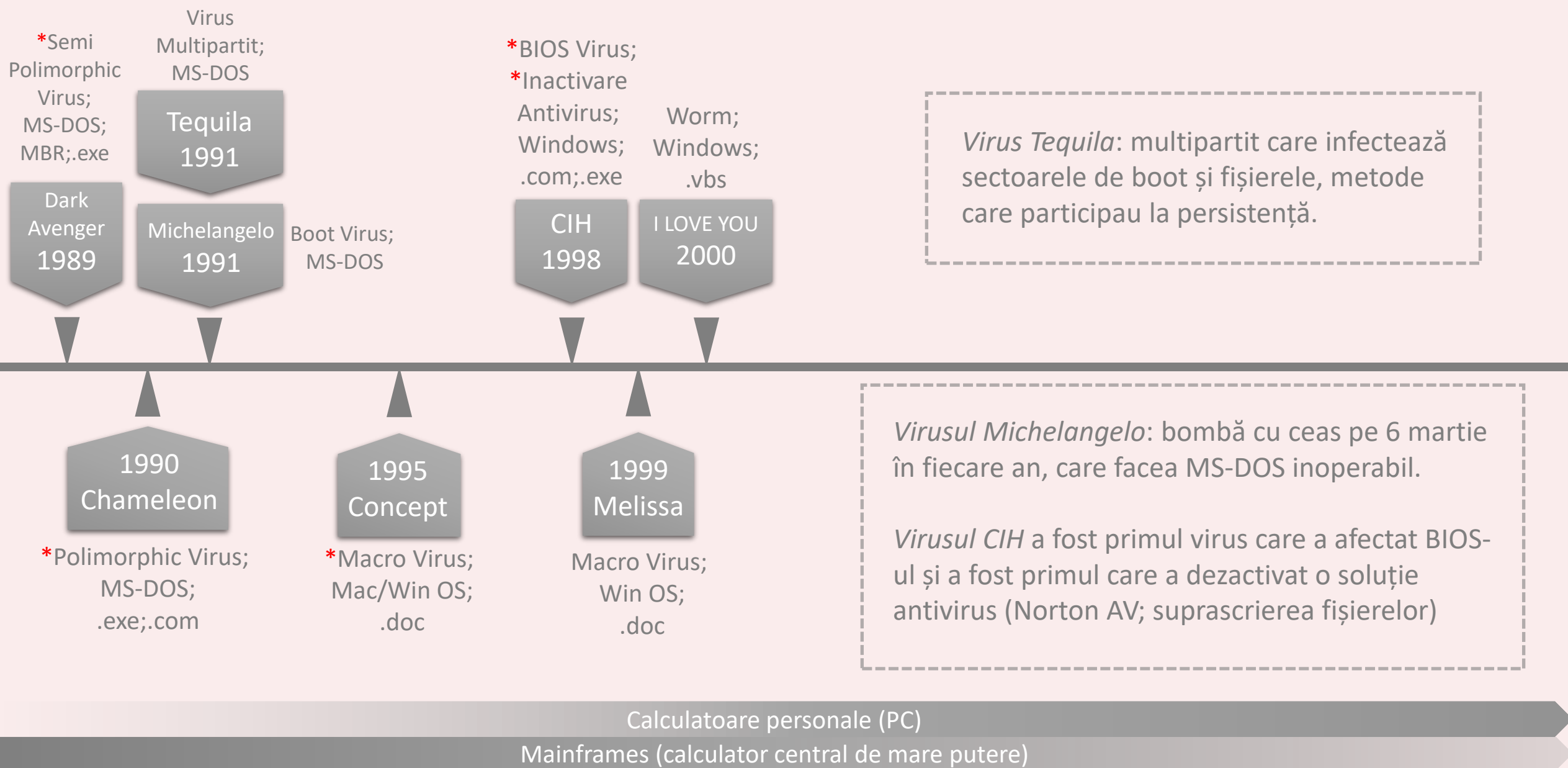
Viermele Creeper: Când se autocopia într-un sistem nou, viermele își ștergea fișierul din vechiul sistem de operare. După infectare, afișa mesajul „Sunt Creeper, prinde-mă dacă poți!”

Calculatoare personale (PC)

Mainframes (calculator central de mare putere)

Explorarea evoluției malware-ului (I)

1970 - 1990



Explorarea evoluției malware-ului (II)

1990 - 2000

Worm;
Windows;
(resident RAM)

Code
Red
2001

Worm;
Windows;
(resident RAM)

Blaster
2003

Worm;
Windows;
(fis. bin)

Conficker
2008

Trojan;
Windows;
.*

Zbot
2010

Code Red era un vierme (worm) care nu avea o componentă de fișier fizic pe disc și era rezident în memorie.

2001
Nimda

2004
Mydoom

2010
Stuxnet

Nimda a fost unul dintre primele virusuri care a combinat caracteristicile unui virus cu cele ale unui vierme.

Copiile lui *Conficker* erau simple fișiere executabile care nu necesitau extensii specifice pentru a fi funcționale.

*Virus/Worm;
Windows;
email
.eml;.exe

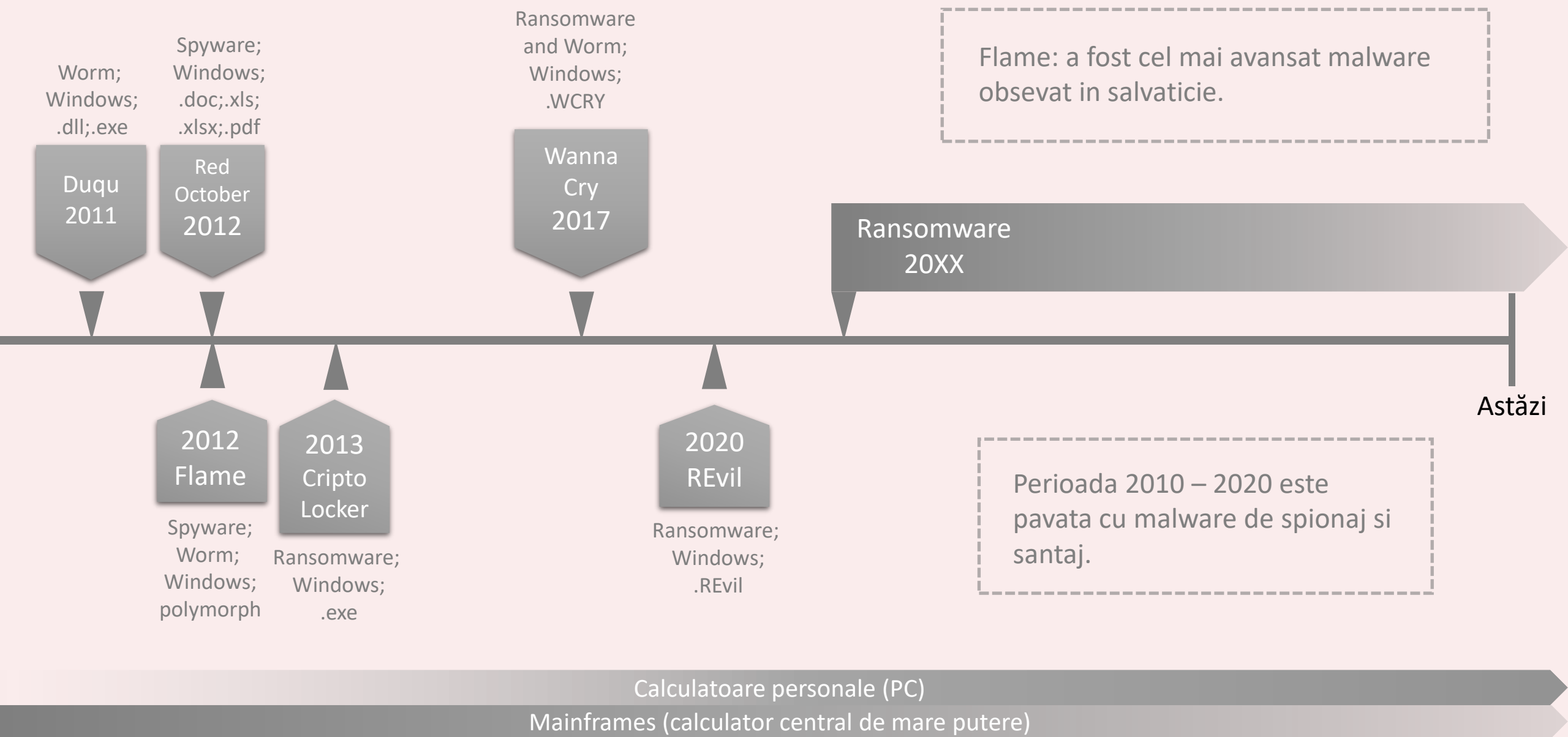
Virus/Worm;
Windows;
email
.exe;.zip;.scr

Worm;
Win OS;
.doc

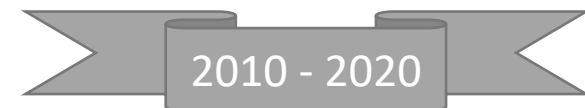
Calculatoare personale (PC)
Mainframes (calculator central de mare putere)

Explorarea evoluției malware-ului (III)

2000 - 2010



Explorarea evoluției malware-ului (VI)



Tipuri de malware/Decada

Distributia aproximativa:

Anii 1970

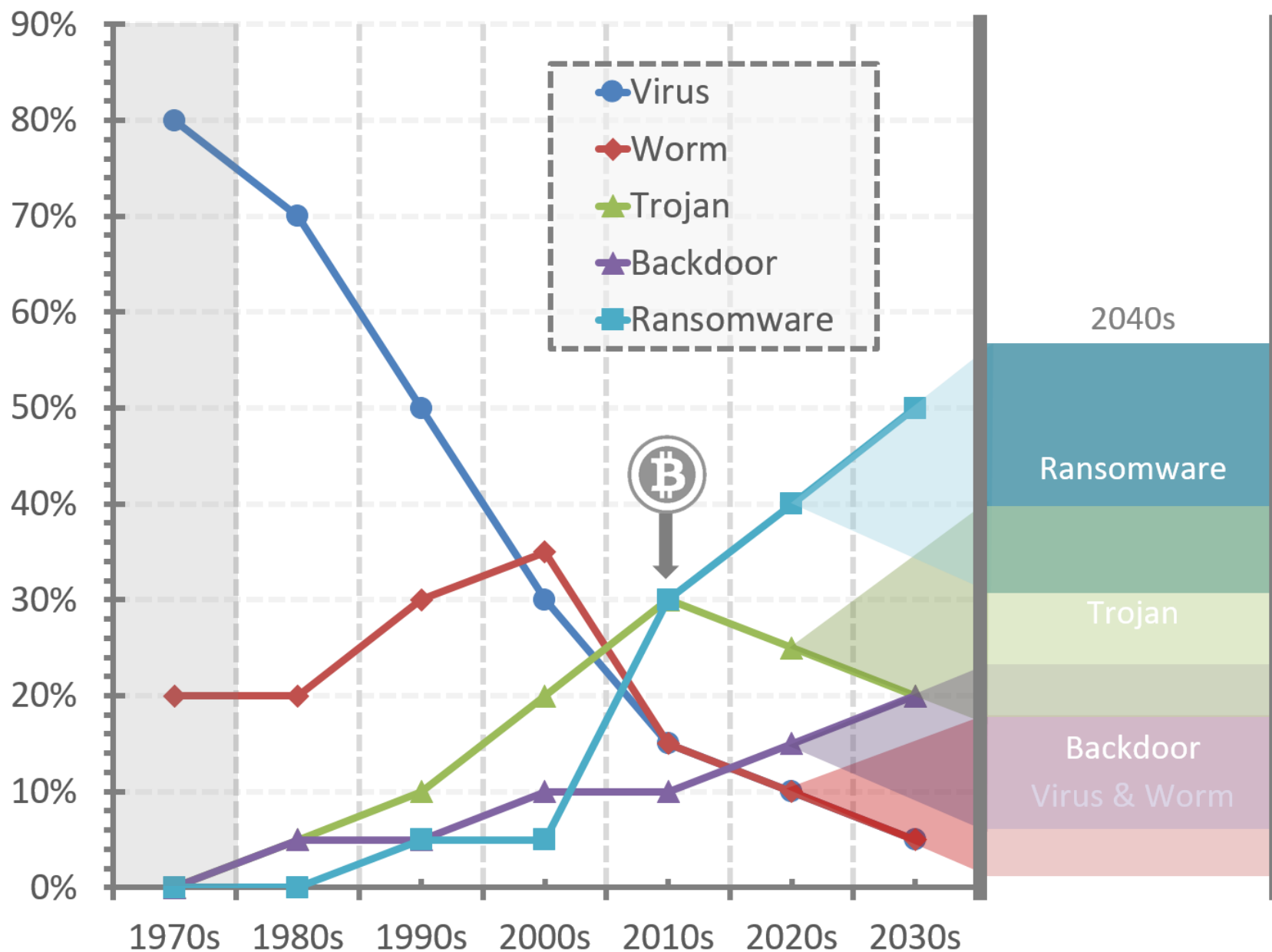
Virus: 80%
(Discuții mai mult teoretice
și dezvoltări timpurii)

Worm: 20%
(Apariția primelor viermi,
cum ar fi *Creeper*)

Trojan: 0%
(Nu a fost semnificativ
recunoscut în acea perioadă)

Backdoor: 0%
(Instance limitate)

Ransomware: 0%
(Nu a fost încă dezvoltat)



Tipuri de malware/Decada

Distributia aproximativa:

Anii 1980

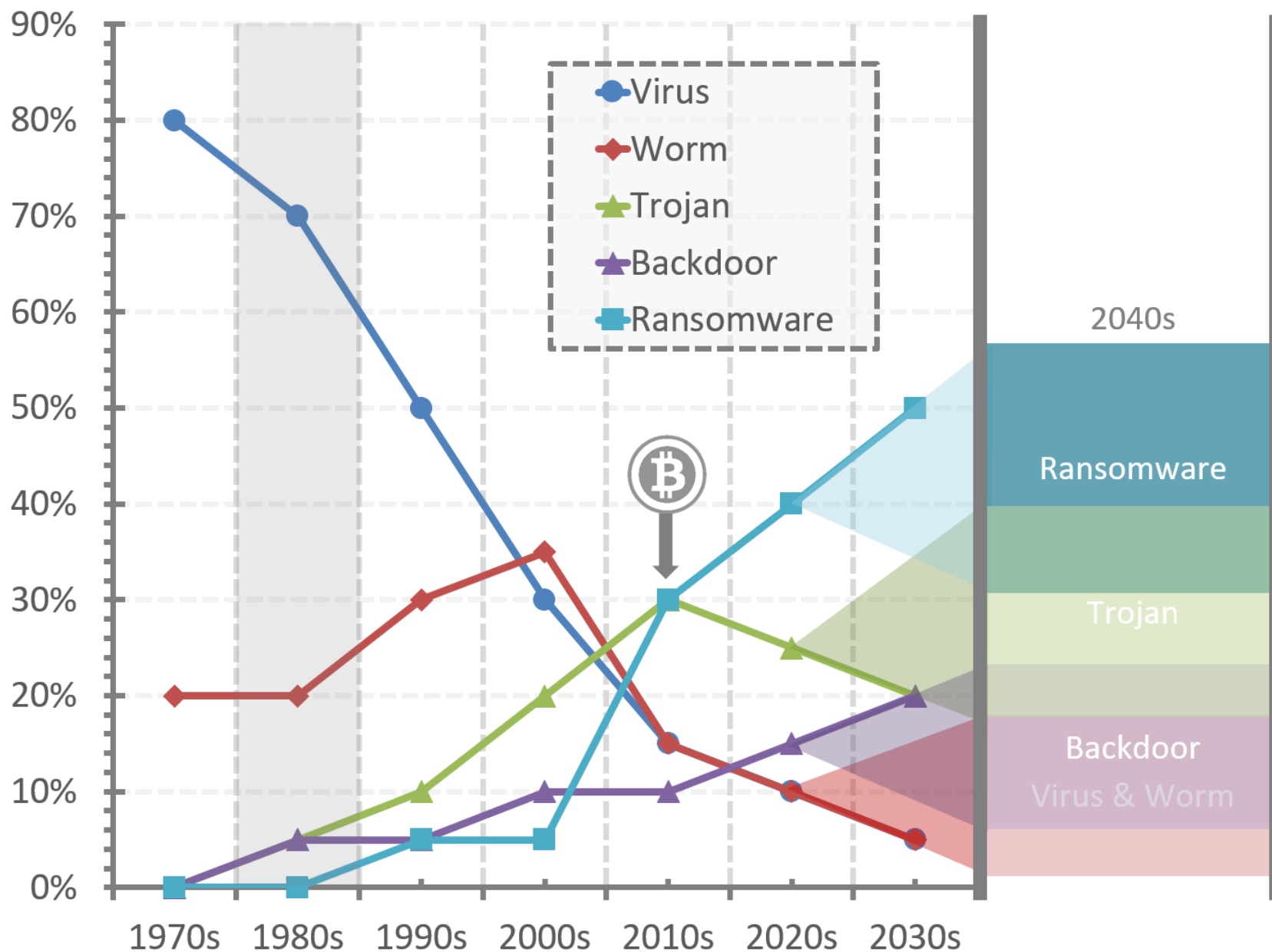
Virus: 70%
(Creșterea virusilor de PC)

Worm: 20%
(Instanțe notabile
precum *Morris Worm*)

Trojan: 5%
(În stadiu incipient,
dar nu proeminent)

Backdoor: 5%
(Începuturile dezvoltării)

Ransomware: 0%
(Nu a reprezentat încă
o amenințare semnificativă)



Tipuri de malware/Decada

Distributia aproximativa:

Anii 1990

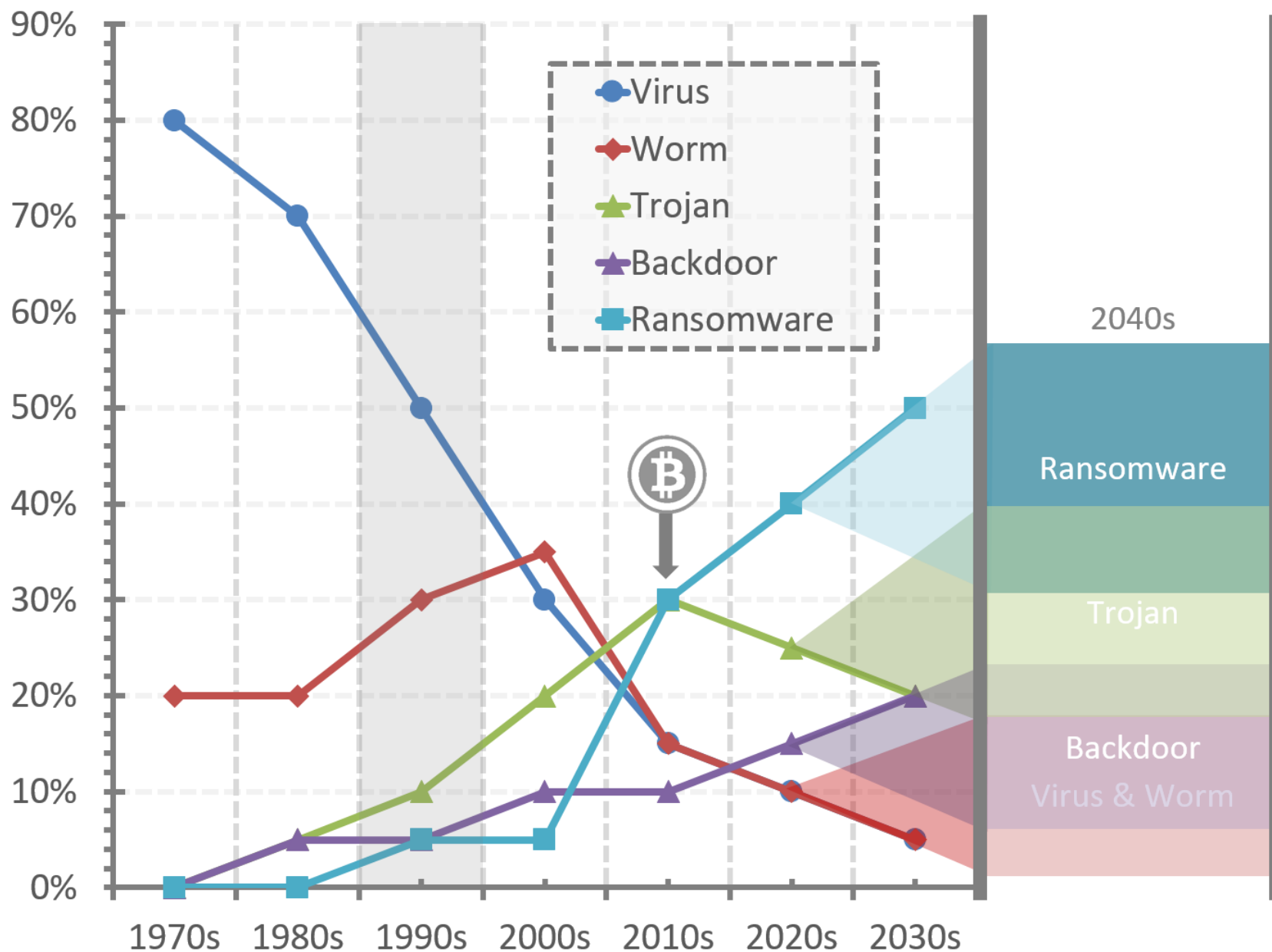
Virus: 50%
(Proliferare larg răspândită)

Worm: 30%
(Activitate crescută,
în special spre sfârșitul anilor '90)

Trojan: 10%
(Începe să câștige teren)

Backdoor: 5%
(Văzut în atacuri mai sofisticate)

Ransomware: 5%
(Formele timpurii încep să apară)



Tipuri de malware/Decada

Distributia aproximativa:

Anii 2000

Virus: 30%
(Încă frecvent, dar umbrit de noi amenințări)

Worm: 35%
(Vârful activităților worm cu focare globale majore)

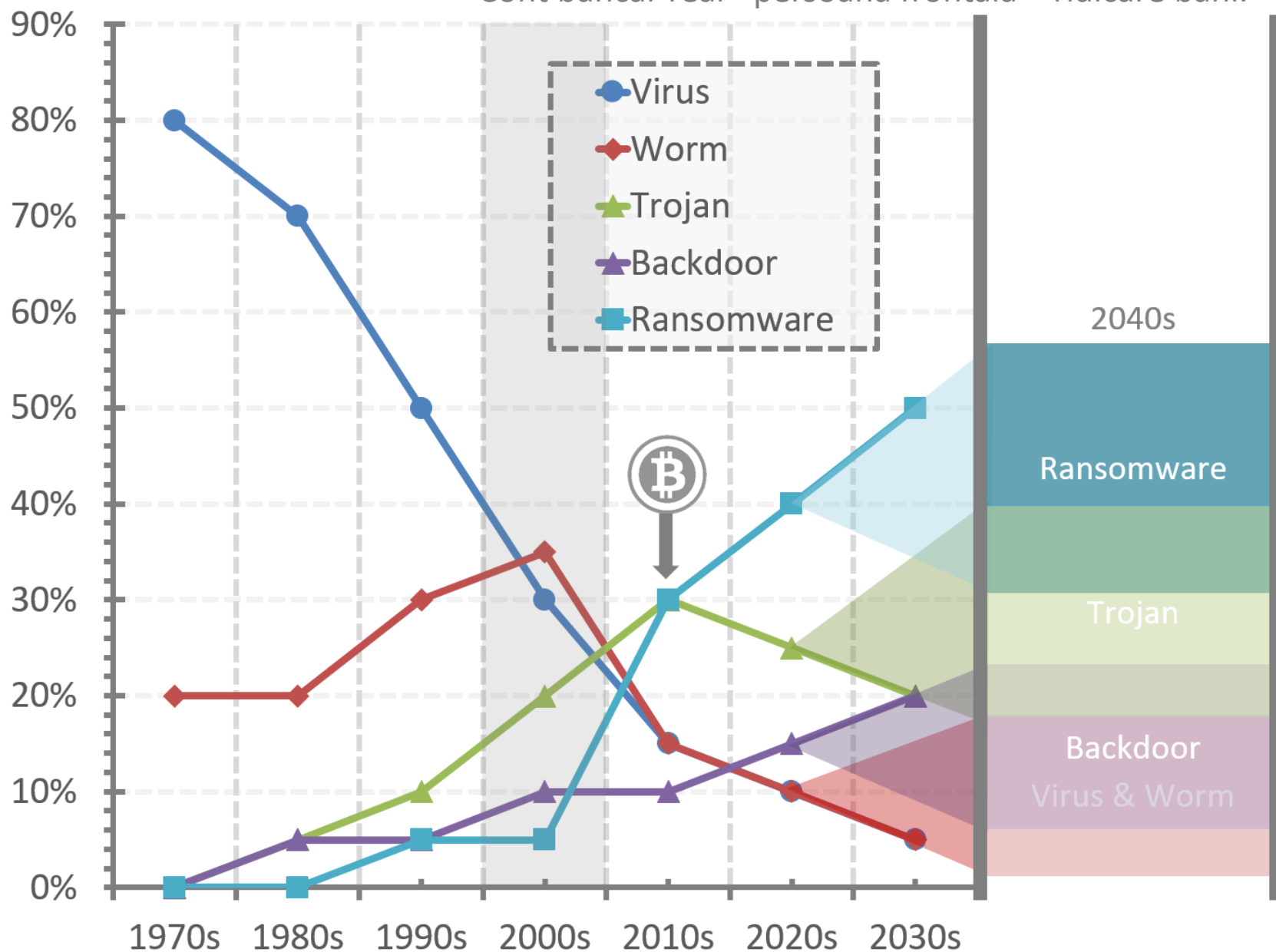
Trojan: 20%
(Creștere semnificativă în sofisticare și prevalență)

Backdoor: 10%
(Utilizat în ciber-spionaj mai complex)

Ransomware: 5%
(Emergența ransomware-ului modern)

Conturi bancare - identitate reală:

- Cumpărare produse – la adresa – vânzare ulterioară.
- Cont bancar real - persoana frontală – ridicare bani.



Tipuri de malware/Decada

Distributia aproximativa:

Anii 2010

Virus: 15%
(Eclipsat de tipuri de malware mai sofisticate)

Worm: 15%
(Încă prezent, dar mai puțin dominant)

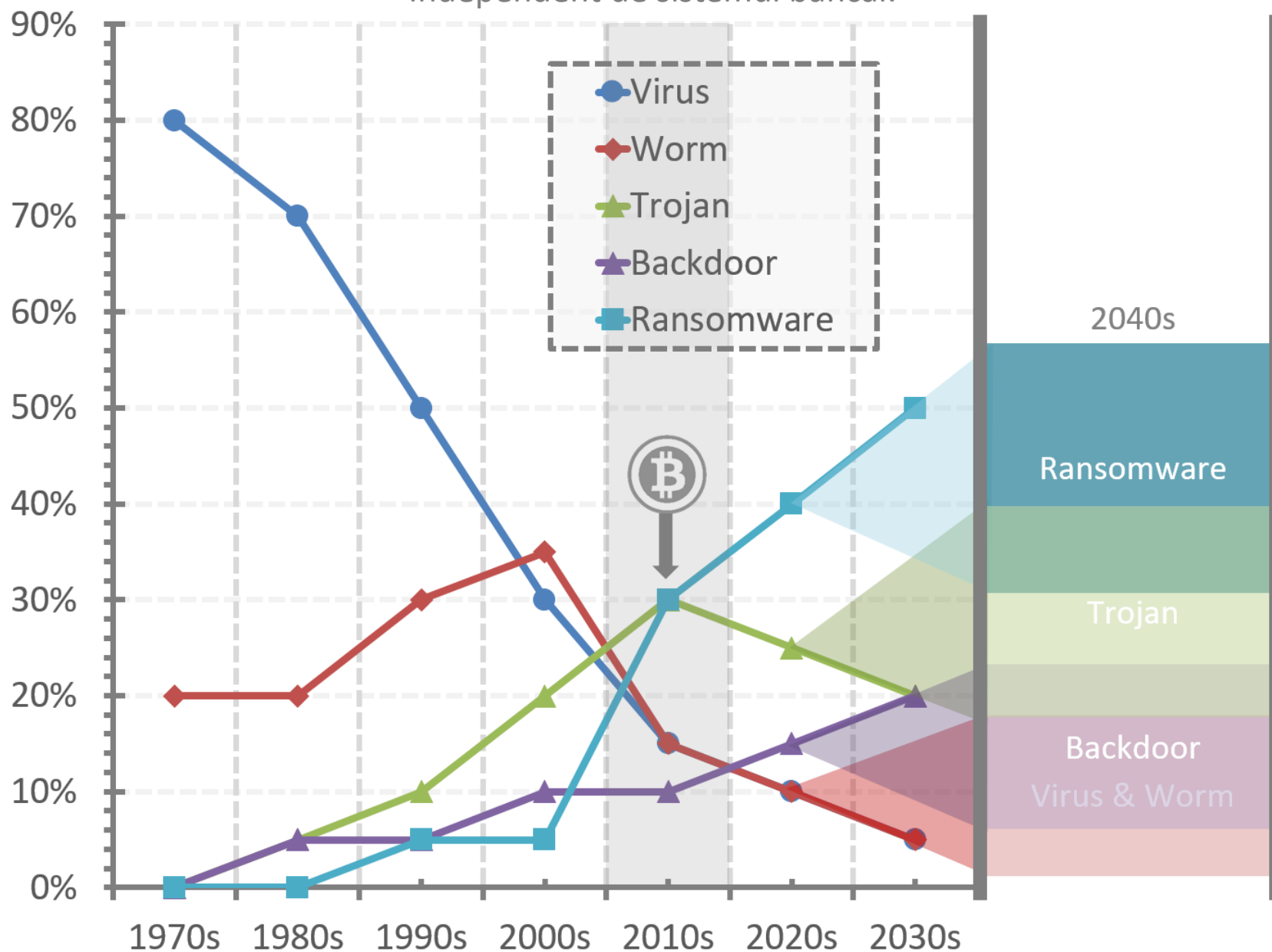
Trojan: 30%
(Semnificativ sub forma APT-urilor și troienilor bancari)

Backdoor: 10%
(În special în atacuri sponsorizate de stat)

Ransomware: 30%
(Creștere rapidă cu atacuri de mare profil)

Apar criptomonezi:

- Incontrolabile – identitate necunoscuta.
- Independent de sistemul bancar.



Tipuri de malware/Decada

Distributia aproximativa:

Anii 2020

Virus: 10%
(Mai puțin comun
în sensul tradițional)

Worm: 10%
(Încă utilizat, dar nu
amenințarea principală)

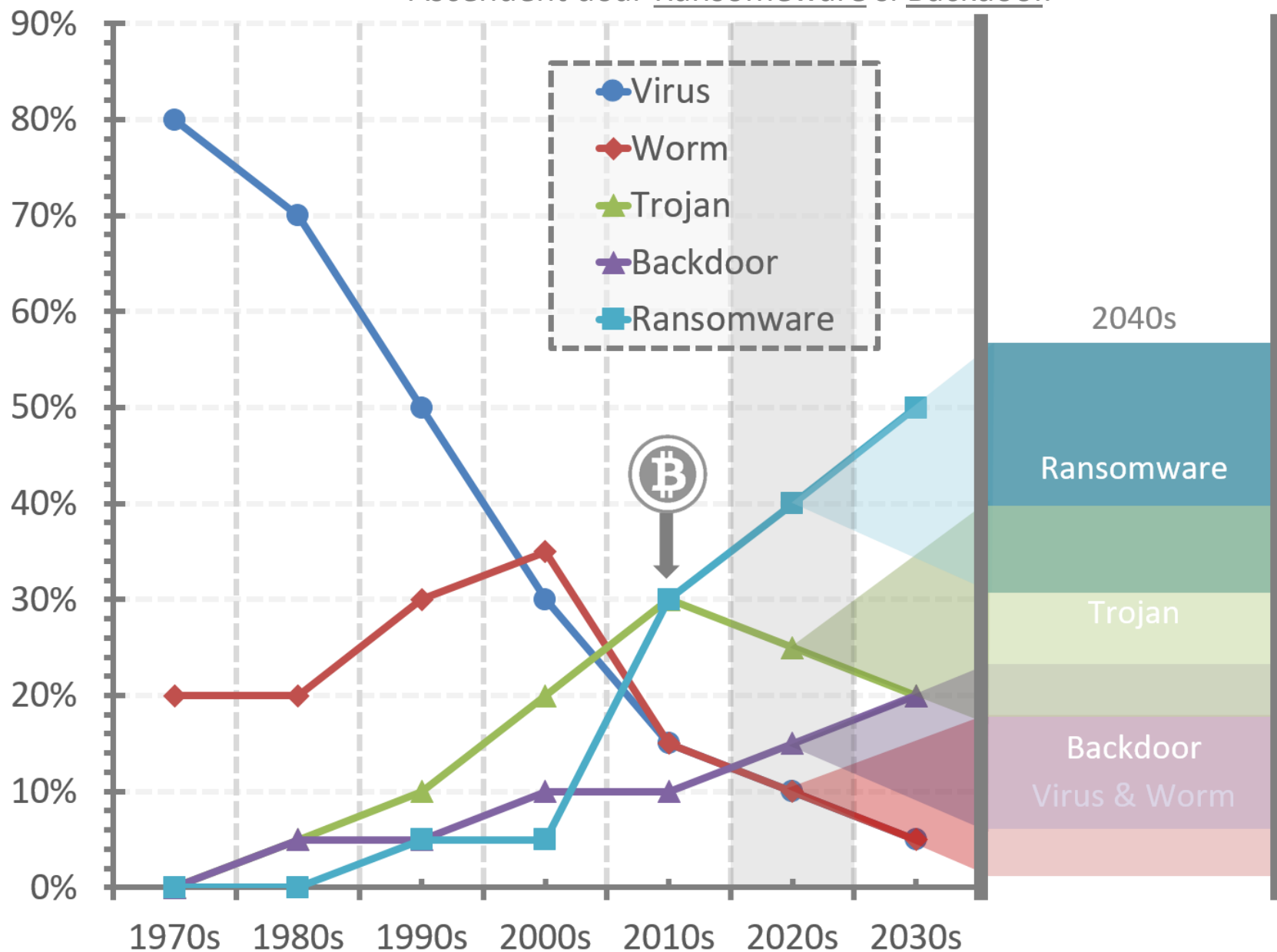
Trojan: 25%
(Continuă să fie important,
mai ales în atacuri țintite)

Backdoor: 15%
(Creștere în sofisticare
și utilizare în spionaj)

Ransomware: 40%
(Forma dominantă de
malware în ultimii ani)

Adaptarea la criptomonezi:

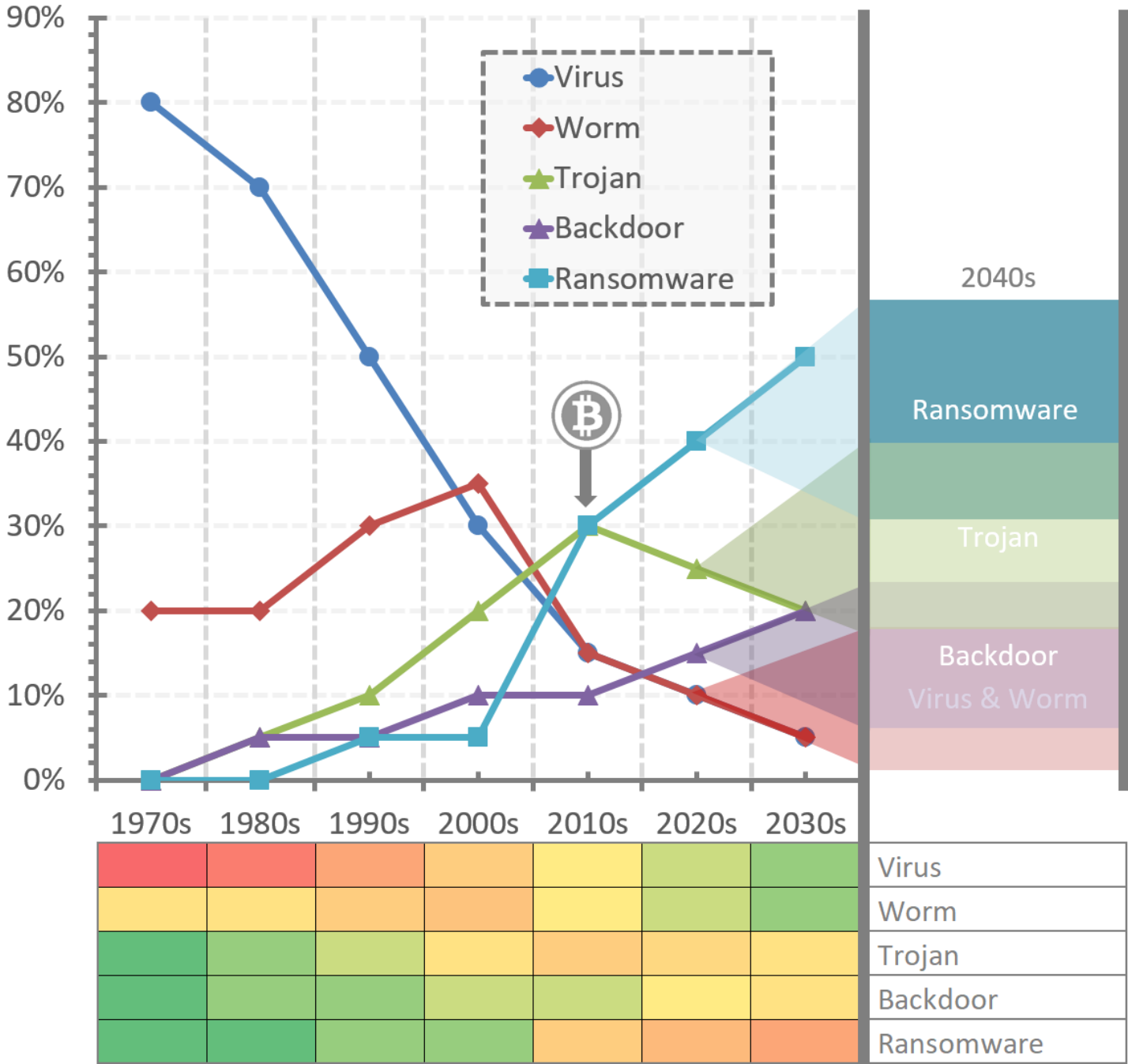
- Reutilizarea malware pentru șantaj.
- Ascendent doar Ransomeware si Backdoor.



Tipuri de malware/Decada

Distributia aproximativa:

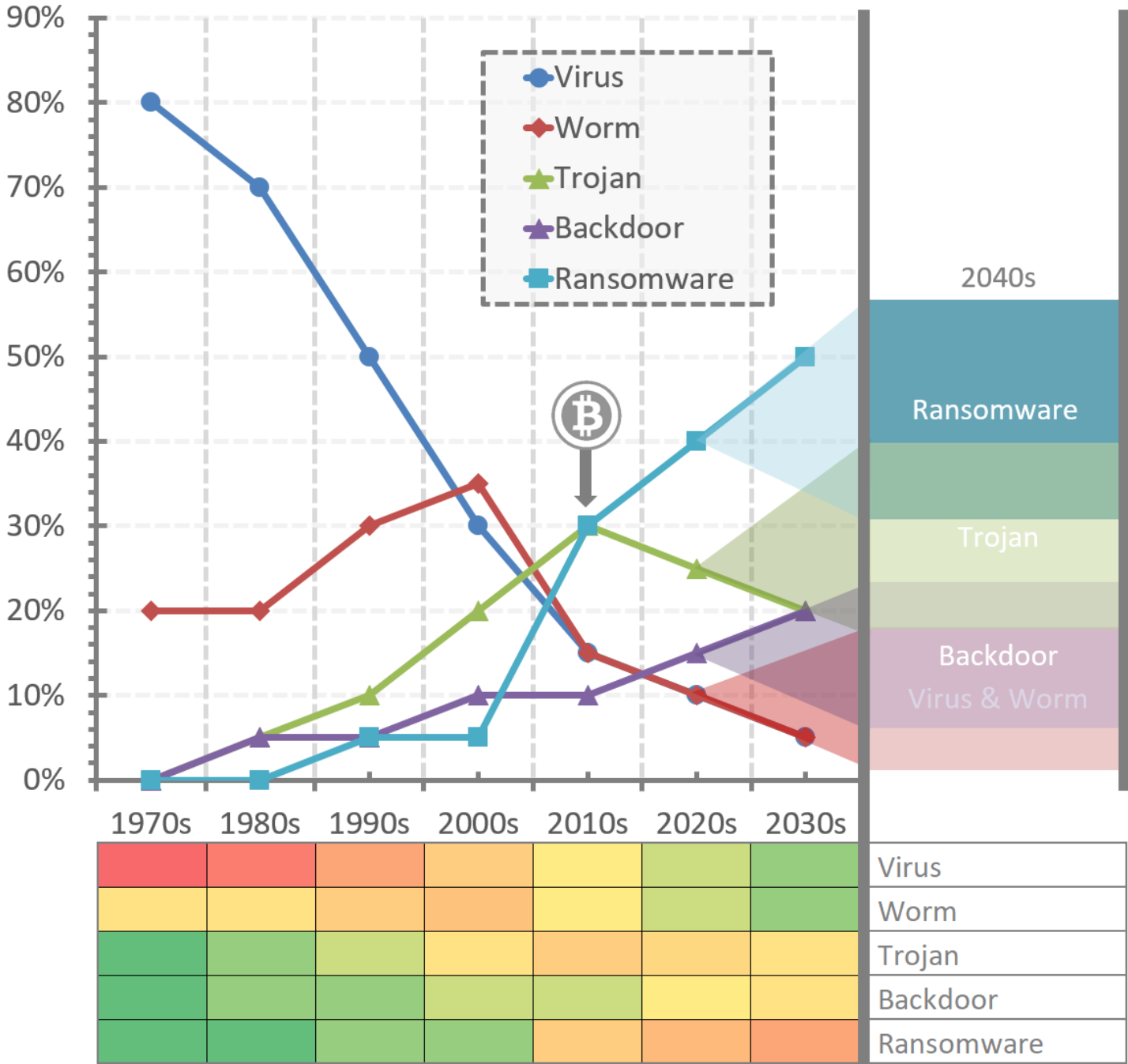
- Aceste procente sunt estimări generale și sunt supuse variațiilor bazate pe diferite surse de date și natura în evoluție a amenințărilor de securitate cibernetică.
- Acestea reprezintă un trend general în peisajul malware, de la virusuri simple auto-replicante la atacuri complexe sponsorizate de stat și ransomware motivat financiar.



Tipuri de malware/Decada

Distributia aproximativa:

- De ce nu mai observam viruși?
 - OS mai vechi au fost eliminate treptat (procese izolate, restricții).
 - Nimeni nu trimite fișiere executabile (email sau USB).
 - Executabile trimise ca arhivă cu parolă (comportamentul forțat).
-
- De ce nu mai observam viermi?
 - Pool-ul de exploit-uri a fost parțial epuizat.
 - Actualizarea OS în timp real închide ușor o potentiala vulnerabilitate.



C.1.3

EVOLUȚIA SOLUȚIILOR DE SECURITATE

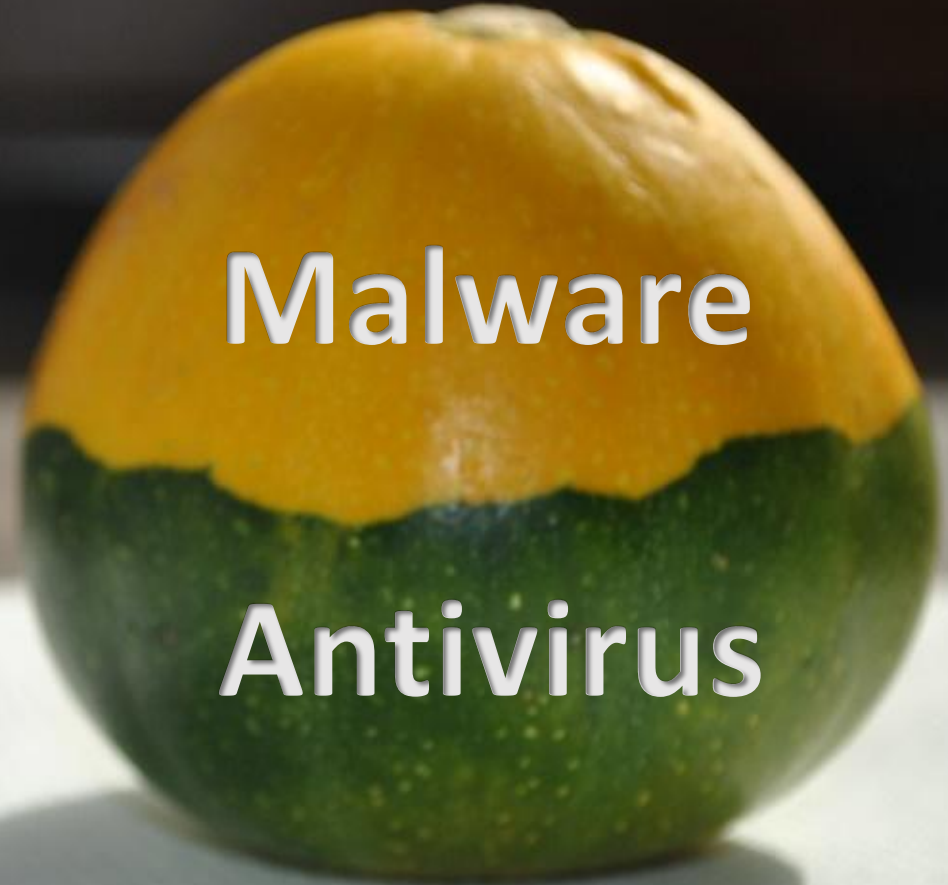


Biologie vs. masini artificiale

Sistem imunitar

Malware

Antivirus



*Unu-la-Unu;
Vs Creaper
(TENEX OS)

Reaper
1971

Uz general;
(MS-DOS)

AVG
Antivirus
1992

Uz general;
(Windows)

Kaspersky
Antivirus
1997

*Al doilea
antivirus
Romanesc

Uz general;
(Windows)

Bitdefender
2001

Uz general;
(Windows)

Malwarebytes
2006

*Al treilea
antivirus
Romanesc

Uz general;
(Windows)

Scut
Antivirus
2008



1988
McAfee
VirusScan

*Primul AV
Comercial
(MS-DOS)

1991
Norton
Antivirus

Uz general;
(Windows)

1994
RAV
Antivirus

*Primul
antivirus
Romanesc

2002
Clam
Antivirus

*Open
source

2009
Windows
Defender

Integrare in Windows

Uz general; (Windows;Linux;UNIX); comparat in 2003

(1969-1990) ARPANET

(1983-present) Internet

Solutii antivirus !

Companiile AV serioase & discriminarea optima.

1970 - 2010



INTERNET ARCHIVE

```
cap    cl, 40h
jbe    7C901B1C
cap    cl, 20h
jbe    7C901B12
shld   edx, eax, cl
shl    eax, cl
ret
mov    edx, eax
xor    eax, eax
and    cl, 1Fh
shl    edx, cl
ret
xor    eax, eax
xor    edx, edx
```

IN HOC SIGNO VINCES



QUIS CUSTODIET IPSOS CUSTODES
CONSILIO ET PRUDENTIA

GLOBAL ADVANCED TECHNOLOGY

SCUT ANTI VIRUS

SCUT FAMILY SYSTEM INFORMATION SECURITY



[Home](#) [Defense Center](#) [World Status](#) [Download](#) [Solutions](#) [eStore](#) [Products](#) [Partners](#) [Virus Encyclopedia](#) [FAQ](#) [About Novus Ordo](#)



Whether you're at home or away, you can rest easy knowing that Scut AntiVirus system is behind you. Antivirus professionals keep a constant, watchful eye on malware, spyware, crimeware, exploits and vulnerabilities that are appearing every day around the globe. The proactive detection of Scut AntiVirus keeps a vigilant eye on your home and your business by constantly monitoring malware or spyware programs.

Scut AntiVirus scans your system for all types of malware, such as viruses, rootkits, trojans, keyloggers, spyware, adware etc. If the threat is detected, it can be easily removed from your computer.

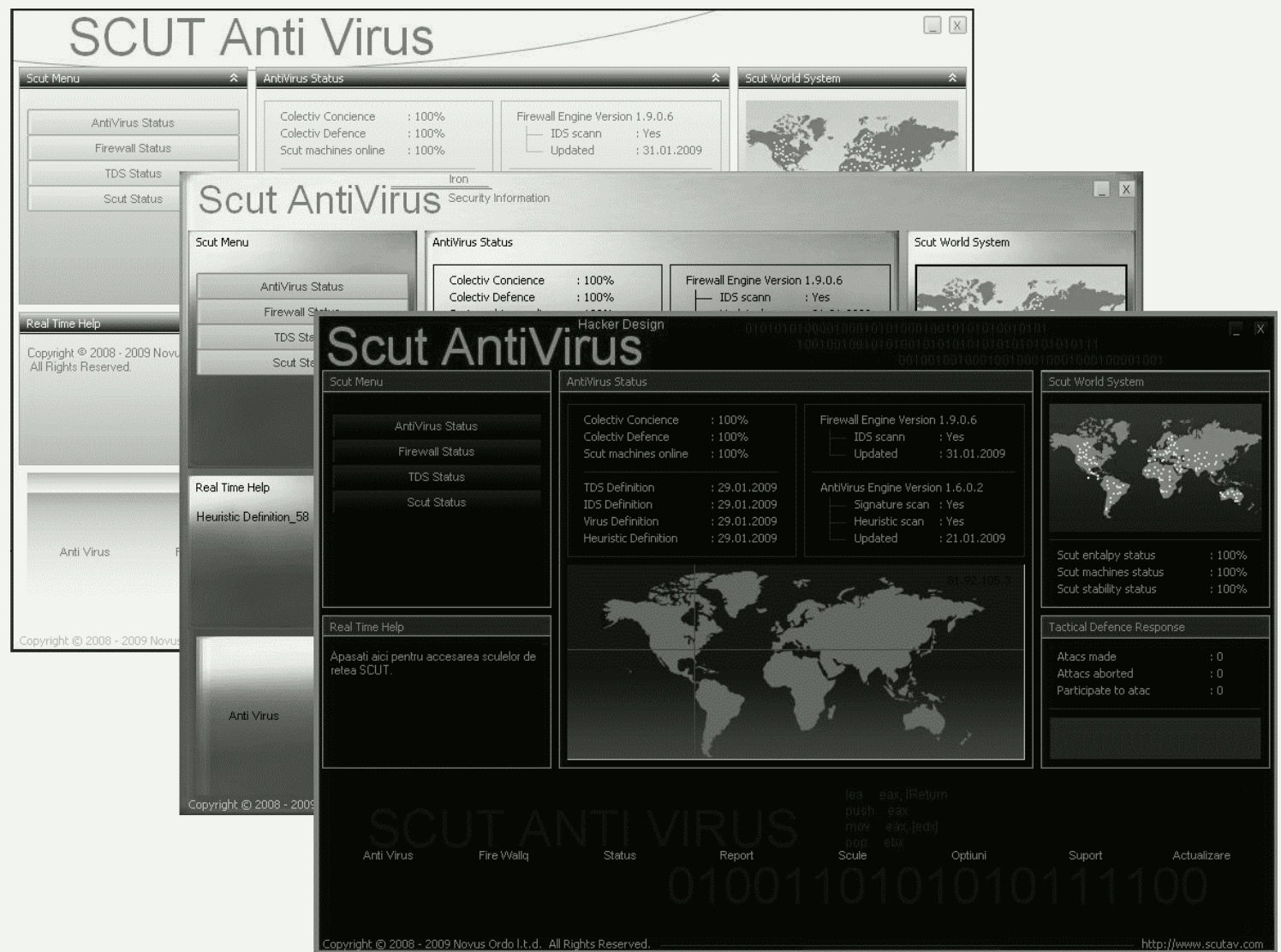
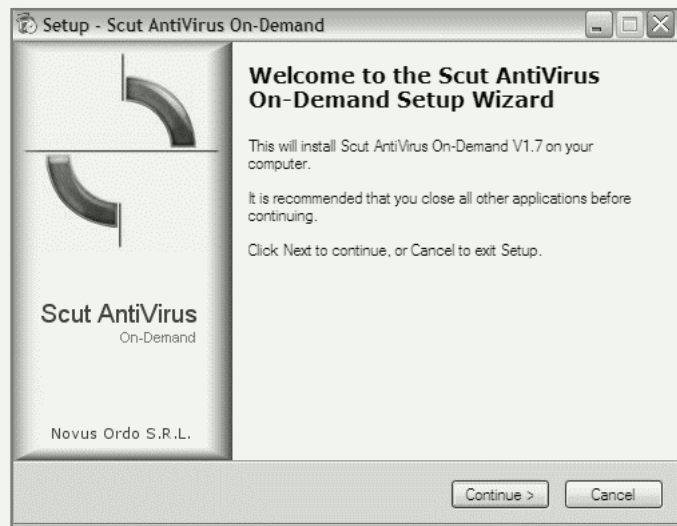
Scut AntiVirus

Total Security



INTERNET ARCHIVE
WayBackMachine

Puțin despre soluția *Scut Antivirus*



VX Heaven – Importanța băncilor malware!

VX Heavens

Library Collection Sources Engines Constructors Simulators Utilities Links Antivirus Forum

Hosted sites

eof-project.net
bi0tic.info
29a
Berniee
Bull Moose
DCA
Doomriderz
IKX
fAMINE
herm1t
hh86
Positron
V-Codez
WarGame

Friendly sites

FreeThinking
Scut Anti Virus
DUSecurity Group
You link here?

Viruses don't harm, ignorance does!

"Everyone has the right to freedom of opinion and expression; this right includes freedom to hold opinions without interference and to seek, receive and impart information and ideas through any media and regardless of frontiers."
Article 19 of "Universal Declaration of Human Rights"

Welcome to VX Heavens! This site is dedicated to providing information about *computer viruses* (or *virii*, as some would prefer) to anyone who is interested in this topic.

This site contains a massive, continuously updated collection of magazines, virus samples, virus sources, polymorphic engines, virus generators, virus writing tutorials, articles, books, news archives etc. Even the viruses for the platforms you've never heard of. We also offer free hosting for virus authors and groups.

Some of you might reasonably say that it is illegal to offer such content on the net. Or that this information can be misused by "malicious people". I only want to ask that person: *"Is ignorance a defence?"*

webmaster@vx.netlux.org

You can help us improve the site by [uploading new stuff](#), [making a donation](#), posting our [link](#) to your site, blog or forum, or [leaving a comment](#). Thank you!

What's new? (January 2010)

28 + LIB/EN: Int13h "The short monologue of virus Wendell from his small 120 megabytes world"
27 + LIB/EN: Adam Reynolds "I.T. IN PRACTICE: Computer viruses"
27 + LIB/EN: Eugene Spafford "Computer Viruses and Ethics"
26 + LIB/EN: Star0 "C to assembly, language point of view"

VX Heaven

Library Collection Sources Engines Constructors Simulators Utilities Links [Donate](#) [Forum](#)

Hosted sites

29a, hell knights crew, Berniee, Bull Moose, DCA, Doomriderz, herm1t, IKX, Positron, RRLF, SPTH

Friendly sites

Scut Anti Virus
Virus News
Virus Collection
Your link here?

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You can help us improve the site by [uploading new stuff](#), [making a donation](#), posting our [link](#) to your site, blog or forum, or [leaving a comment](#). Thank you!

✉ webmaster@vxheaven.org

STOP RUSSIAN AGGRESSION AGAINST UKRAINE!

What's new? (March 2016)

No news is a good news :-)

Latest discussions on forum:

CaptainTech in [Can a virus actually damage PC hardware?](#) (2016-03-28 23:14:04)
Xylitol in [http://www.9supplements.org/1k-daily-profit/](#) (2016-03-28 22:43:51)
alcopaul in [\[Q\] Meaning of numbers in the virus alias](#) (2016-03-28 21:36:14)
slek in [http://www.9supplements.org/1k-daily-profit/](#) (2016-03-28 21:02:13)



The screenshot shows the Scut antivirus application window. The title bar reads "Scut antivirus". The main area has several sections:

- In acest modul se introduc automat doar fisierele virus curate, semnăturile luate din fisierele infectate se editează manual!**
- Director banca malware**: Includes fields for "Cale" (Path) and "Masca" (Mask), and a "Scan" button.
- Cautare optiuni**: A section for search options with checkboxes for "Normal", "Arhivat", "Ascuns", "Citire numai", "System", "Pattern Matching", "Cautare in sub-directoare", "Sorteaza fisiere (A-Z)", "Reimprospatare ListView", and "Use Extended Mode Scan".
- Filtrare data/dimensiune**: Fields for "Dupa data" (After date), "De la" (From), "La" (To), and "Dim fisier" (File size).
- Geneaza siemnatURI...**: A section for generating signatures with a "Stop" button.
- Statusul**: Displays statistics like "Status (%)", "Nr. fis virale:", "Repetitii (%)", "Coincidenta (%)", "Dimensiune sig:", and "Incerari extr.: 6".
- DB virusi !**: A section for managing the virus database with buttons for "Deschide DB >", "Salveaza manual", "Update manual", "Sterge inregistrarea", "Sterge DB-ul", and "File Size 3.14 Kb".
- Semnături**: A table showing detected signatures with columns for "FieldCount", "Lungime semnatura", and "Sig". It includes buttons for "<<", "<", ">", ">>".
- Gaseste semnatura sau nume malware !**: A search field for malware signatures or names.
- BINAR - Virusi din care nu s-a extras nimic: [17]**: A list of binaries from which no virus was extracted.
- BINAR - Semnatura mai mica de 20: [17]**: A list of binaries with a signature smaller than 20.
- Coincidente**: A section for coincident results with a "Text:" field and a "Kappa" value.
- Rezultate :**: A table showing results with columns for "Offset", "IC", and "Coincidente :".
- Coincidenta in jur de : 0.44%**: A summary statistic.
- Distanta fata de sigl:50**: A field for distance from signature length.
- Admis coincidenta: 50**: A field for admission coincidence.
- Admis nr caractere: 10**: A field for admission number of characters.
- Createaza DB si INDEX** and **sorteaza** buttons at the bottom.
- RecordCount > 16** at the very bottom.

The screenshot displays the Semnatura application window, which is used for malware analysis. The interface is in Romanian and includes several sections for configuration and results.

Director banca malware: This section shows the file path being analyzed: `C:\Users\Paul\Desktop\banca\vir`. The file size is listed as 1315 bytes.

Cautare optiuni: This section contains checkboxes for various search options. The checked options are: Normal, Arhivat, Ascuns, Citire numai, and System.

Optiuni: This section contains checkboxes for additional options. The checked options are: Pattern Matching, Cautare in sub-directoare, Sorteaza fisiere (A-Z), Reimprospatare ListView, and Use Extended Mode Scan.

Filtrare data/dimensiune: This section allows filtering by date and size. The date filter is set to "Modified" and the size filter is set to "Any Size".

Scan: A button labeled "Scan" is present, indicating the start of the analysis process.

Malware Analysis Results: The results are displayed in a table-like format. The file being analyzed is `Trojan-PSW.Win32.OnLineGames.arw` located at `C:\Users\Paul\Desktop\banca\vir\`. The analysis shows the following details:

- Malware Name:** Backdoor.Win32.Popwin.asc
- Malware Type:** Backdoor
- Content Type:** Binary
- Malware Size:** 20846 bytes
- Danger Level:** High

The MD5 hash is `02bb2879df4e90bc64c6fbdc0c7785`.

First Entry Point bytes: The first entry point bytes are `9CE8F6FFFFFF9DEBF1C2CC34FEFFFF909060A40EB3C8DB55CFEFFFF8B0683F8010F844B020000C706010000008BD58B5F0DFFFF2BD08995F0DFFFF019520`.

Entry Point: The entry point is `0000B49E`, the linker version is `6.0`, the sub-system is `Windows GUI`, the file offset is `0000069E`, the virtual address is `0000B000`, and the pointer to raw data is `00000200`.

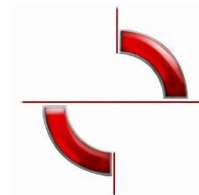
Functions used by this malware: The functions used by the malware are listed in a table:

Function Name	Function Role
GetProcAddress	DLL
LoadLibrary	DLL
ExitProcess	PROCESSES
OpenProcess	PROCESSES
VirtualAlloc	RAM
VirtualFree	RAM
VirtualFreeEx	RAM
VirtualProtect	RAM

The application also features an "Encrypt" button and a "none" option for encryption.

VIRUS ENCICLOPEDIA

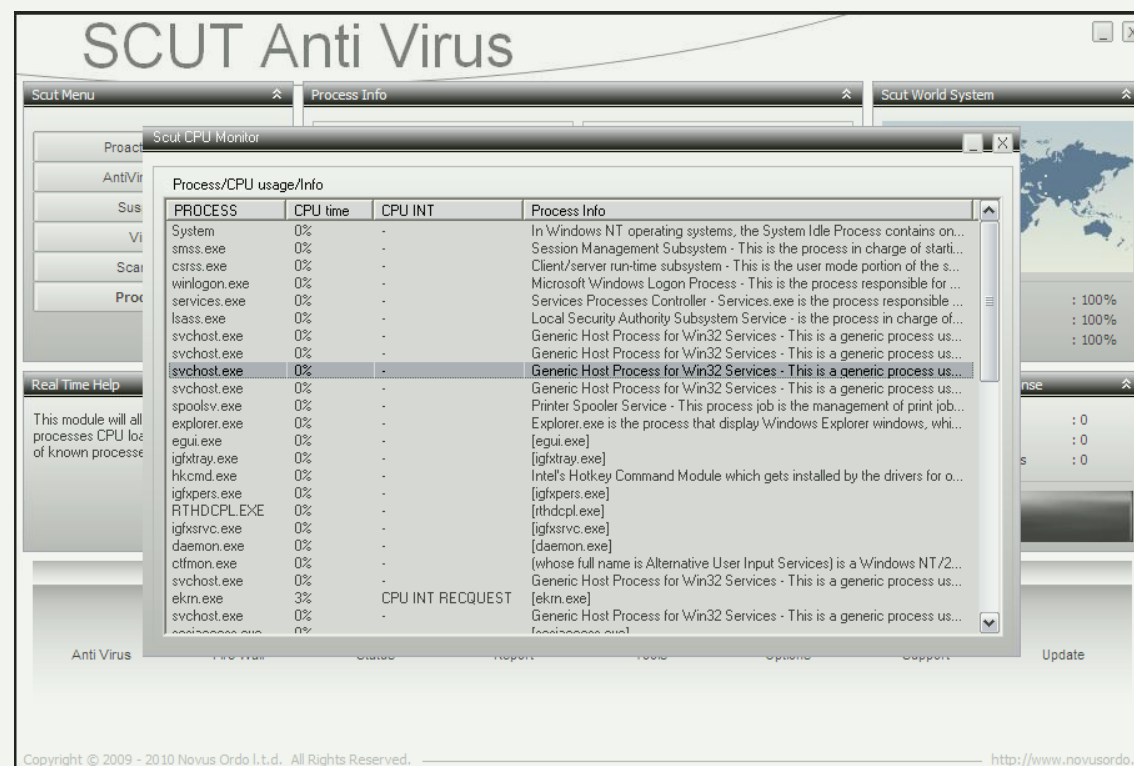
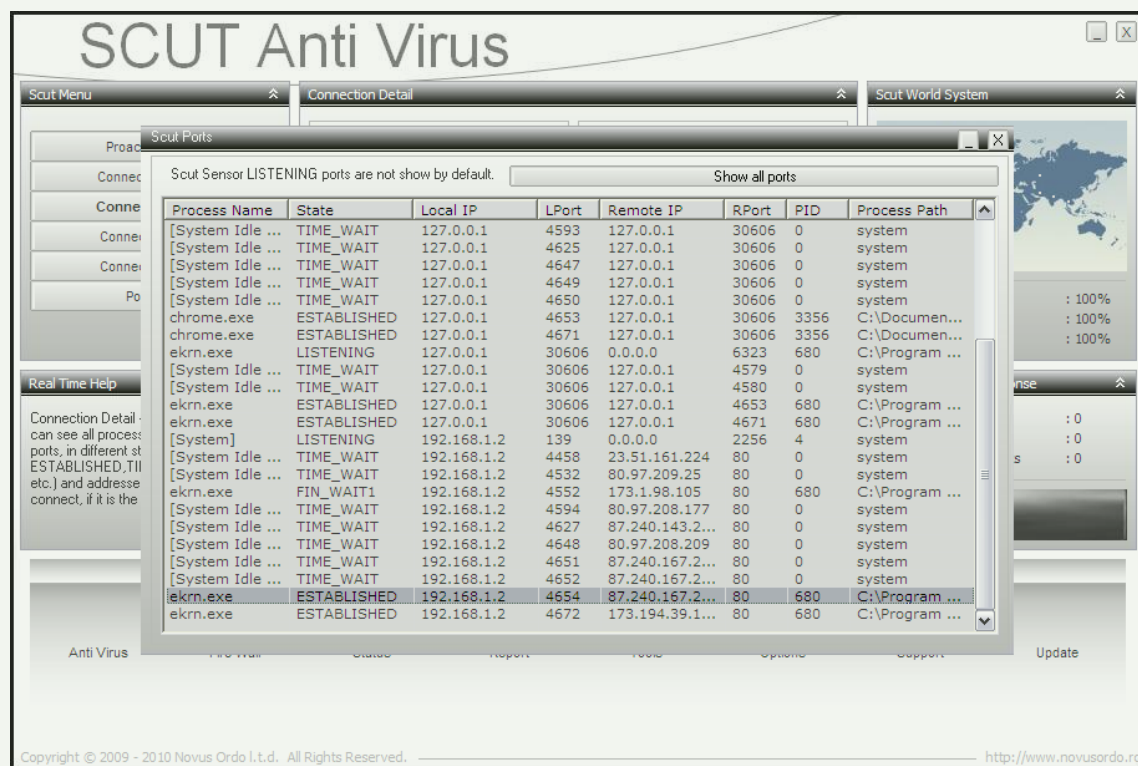
Puțin despre soluția *Scut Antivirus*



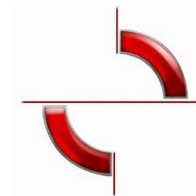
Termenul antivirus a prins rădăcini încă din anii 1980, când tipul majoritar de malware era virusul.

Scut AntiVirus

Astăzi acest termen (i.e. “antivirus”) este încă folosit datorită respectului cuvenit experților în securitate care au fost pionierii sistemului imunitar artificial pe care îl avem astăzi.



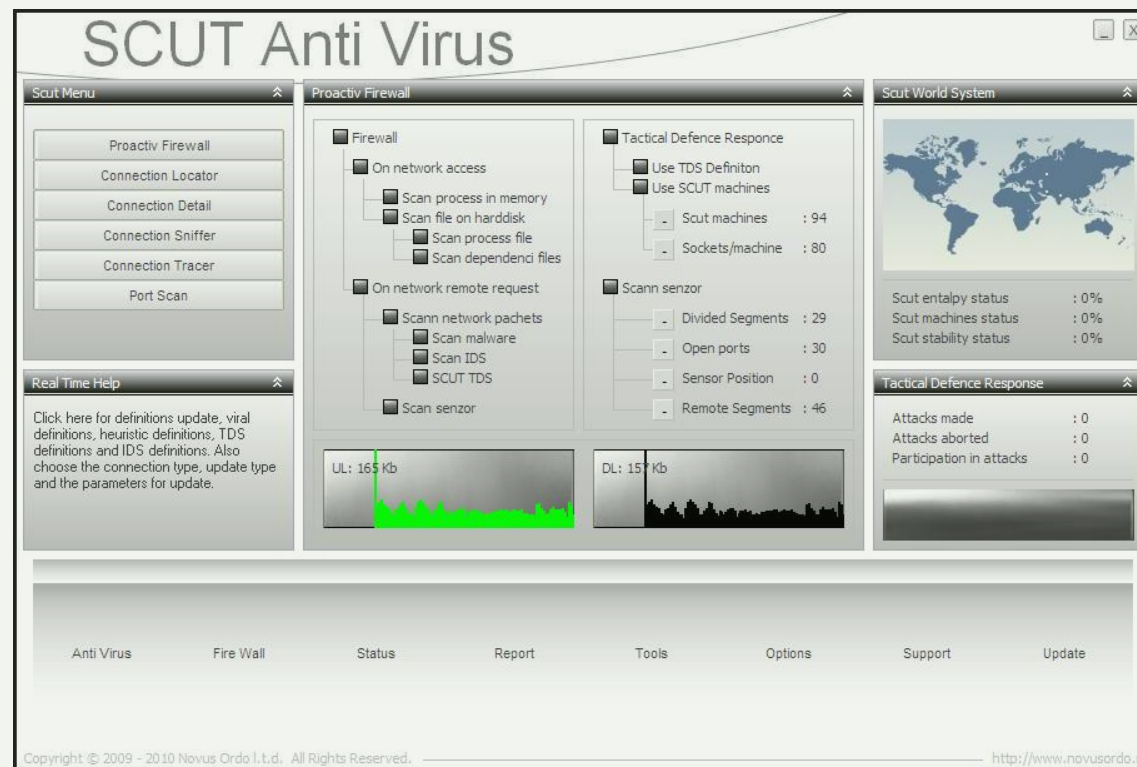
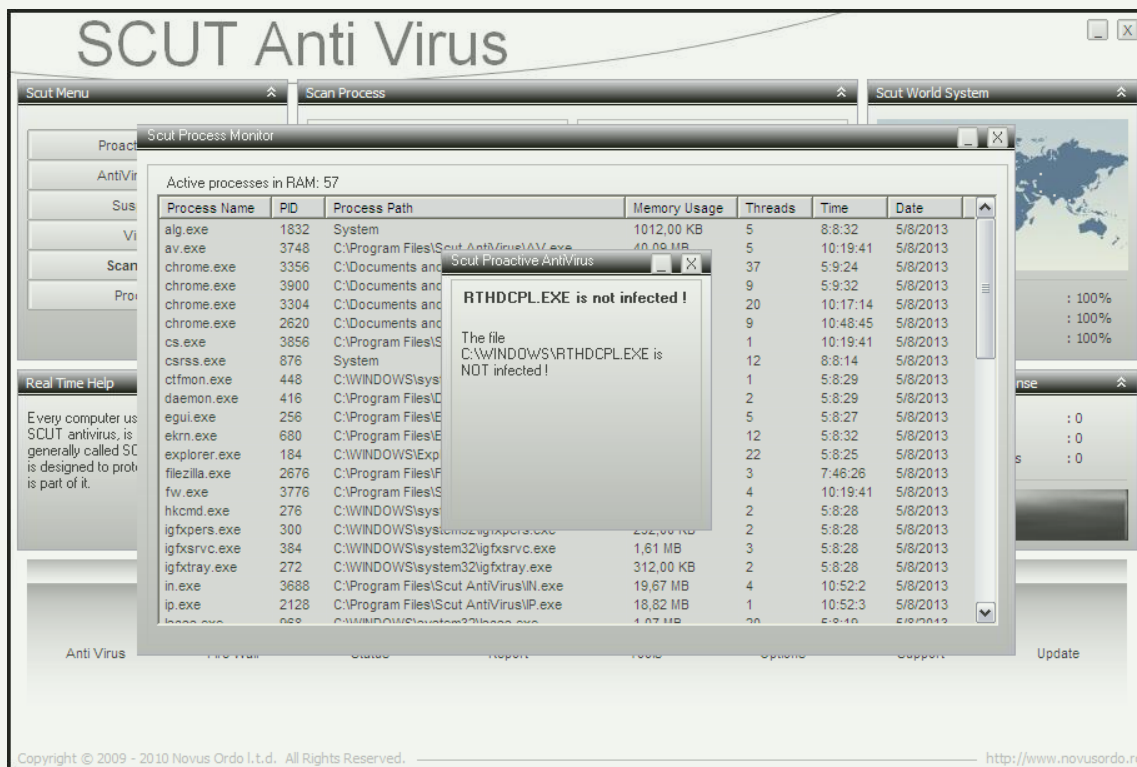
Puțin despre soluția *Scut Antivirus*

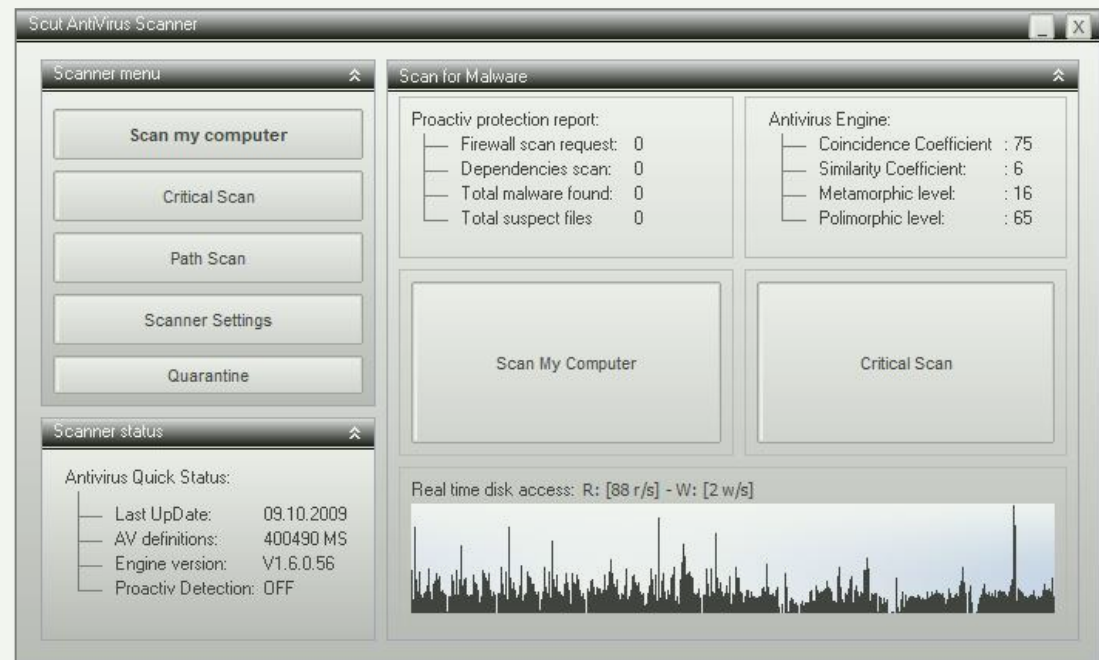
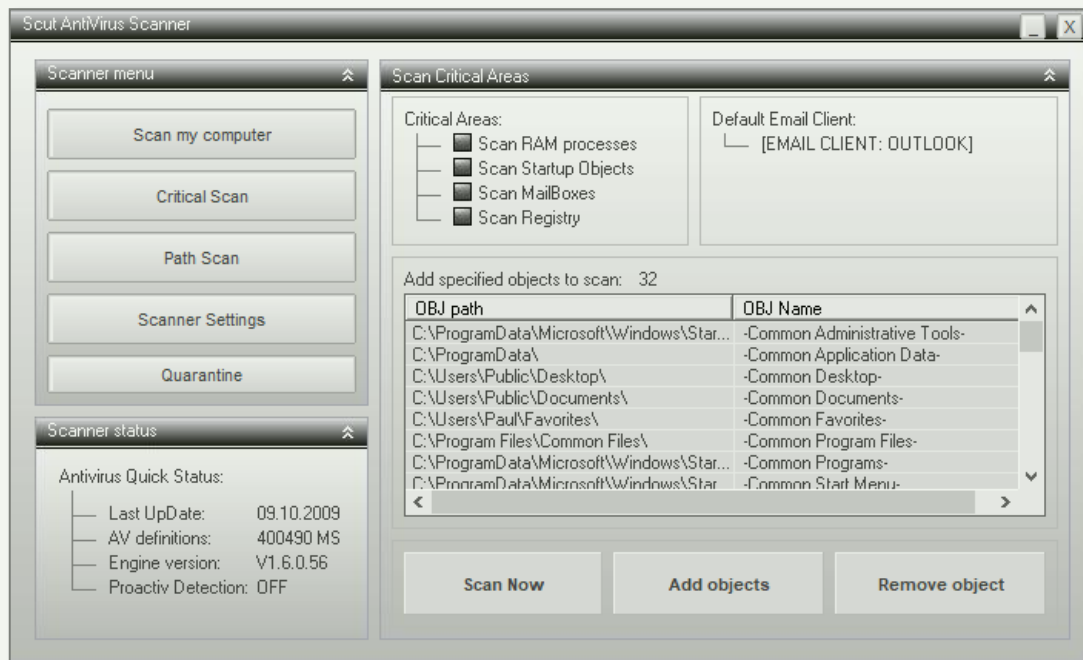
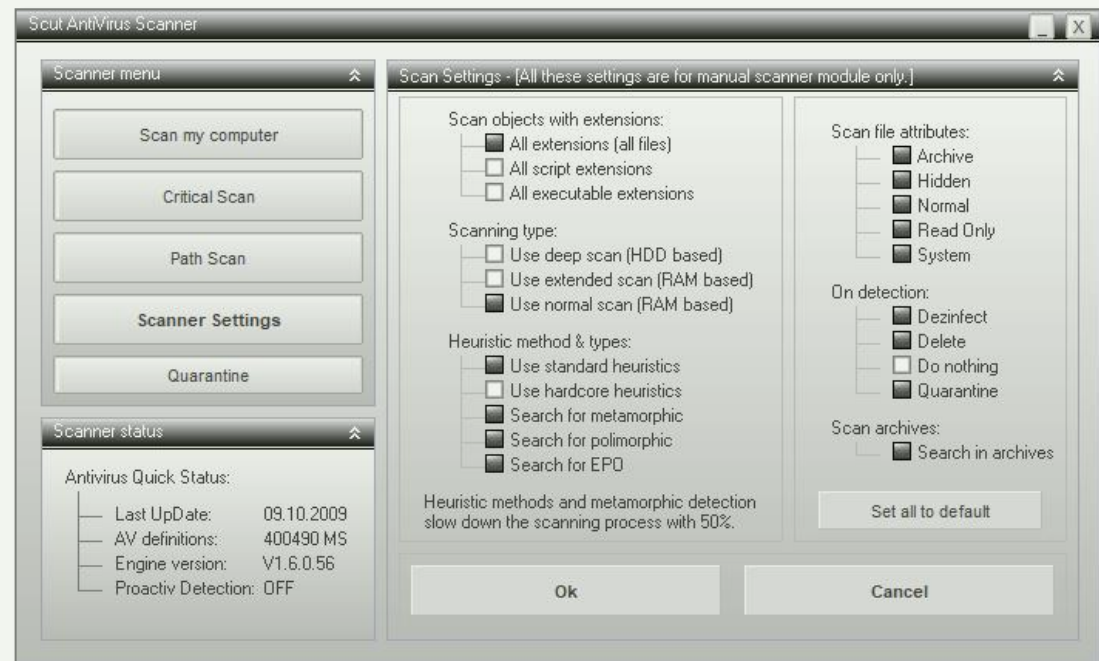
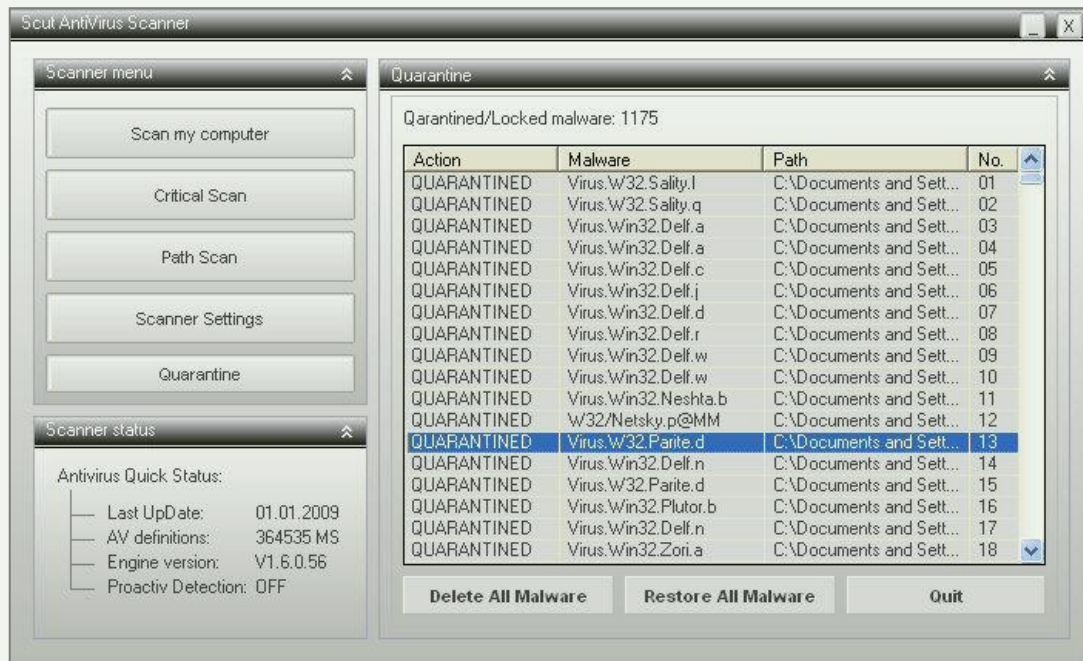


Scut AntiVirus

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Puțin despre soluția *Scut Antivirus*

Scut AntiVirus



ChaaaatTTTTTTTTTTTT100100011001010010101 ...
GPTTTTTTTTTT010011000101001001001 ...

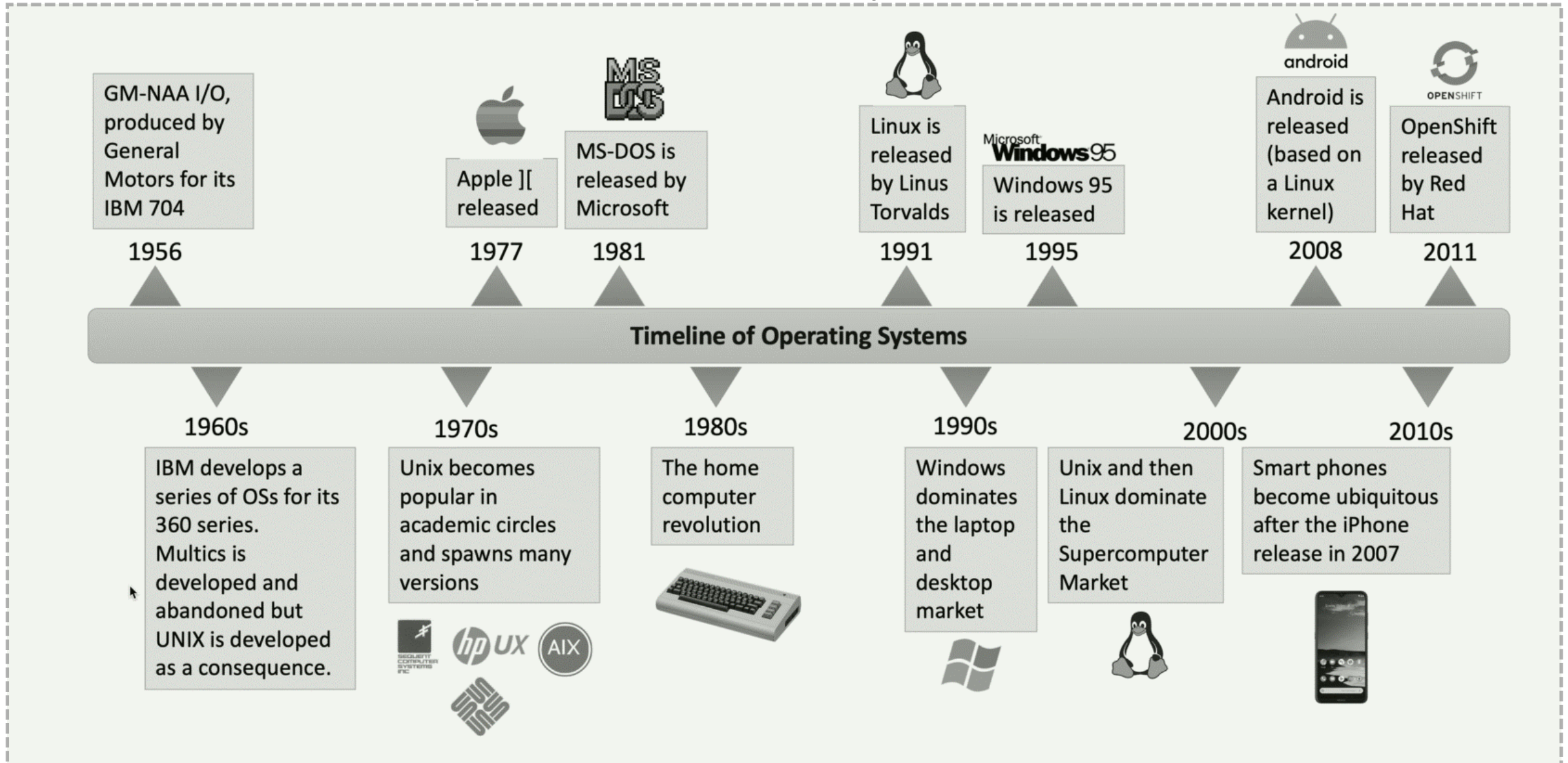


C.1.4

SISTEMELE DE OPERARE ȘI STRATUL DE PORTABILITATE

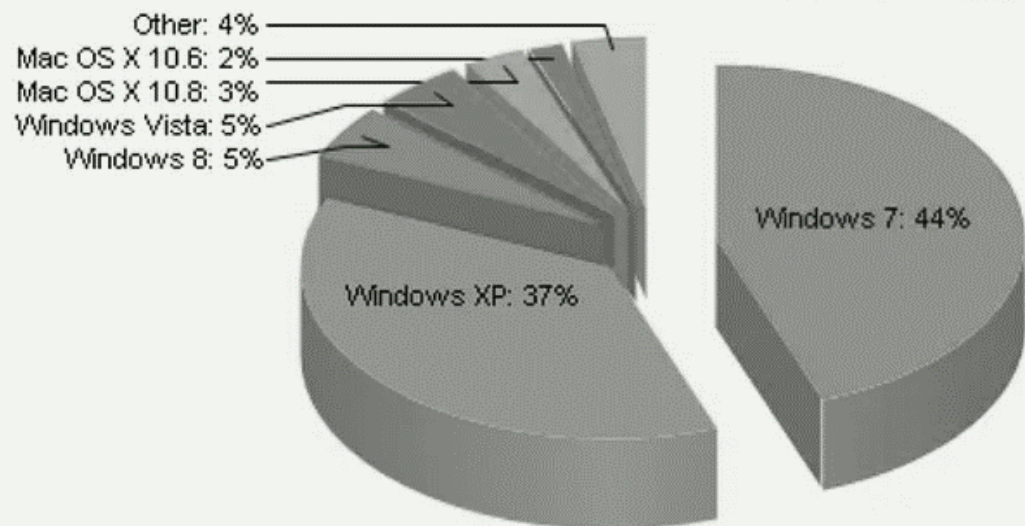
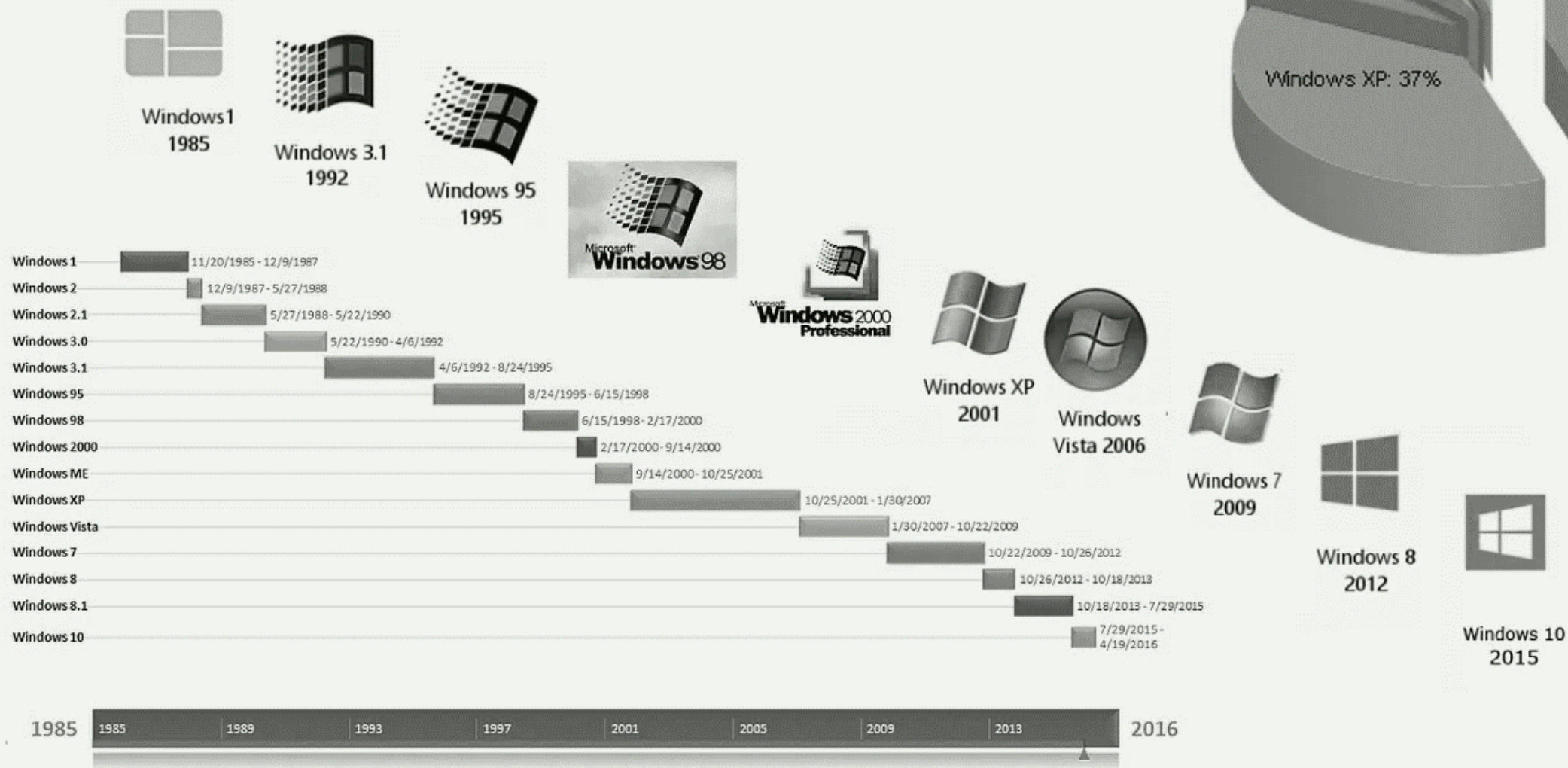


Sistemele de operare în timp



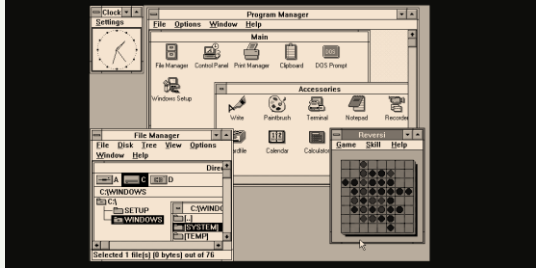
Windows Operating Systems

VERSION RELEASE DATE AND LIFECYCLE



Structura OS a dictat varianta de malware

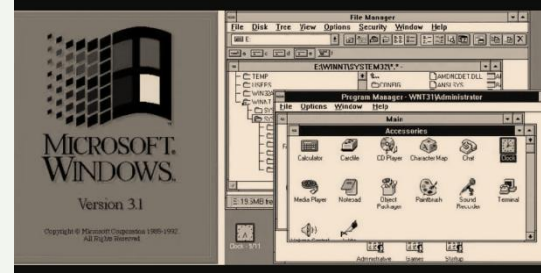
Windows 1



Windows 2.0



Windows 3.1



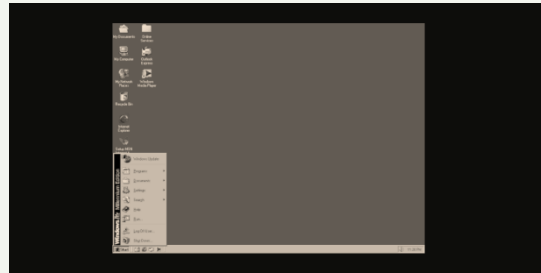
Windows 95



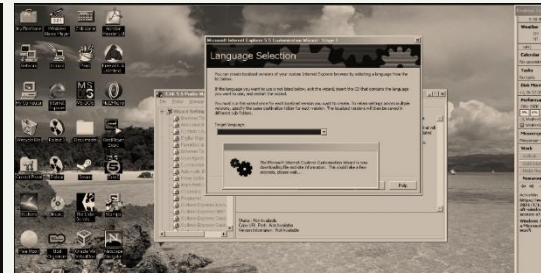
Windows 98



Windows 2000



Windows NT



Windows XP



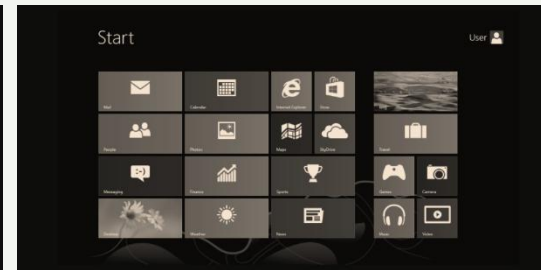
Windows 7



Windows Vista



Windows 8.0



Windows 10

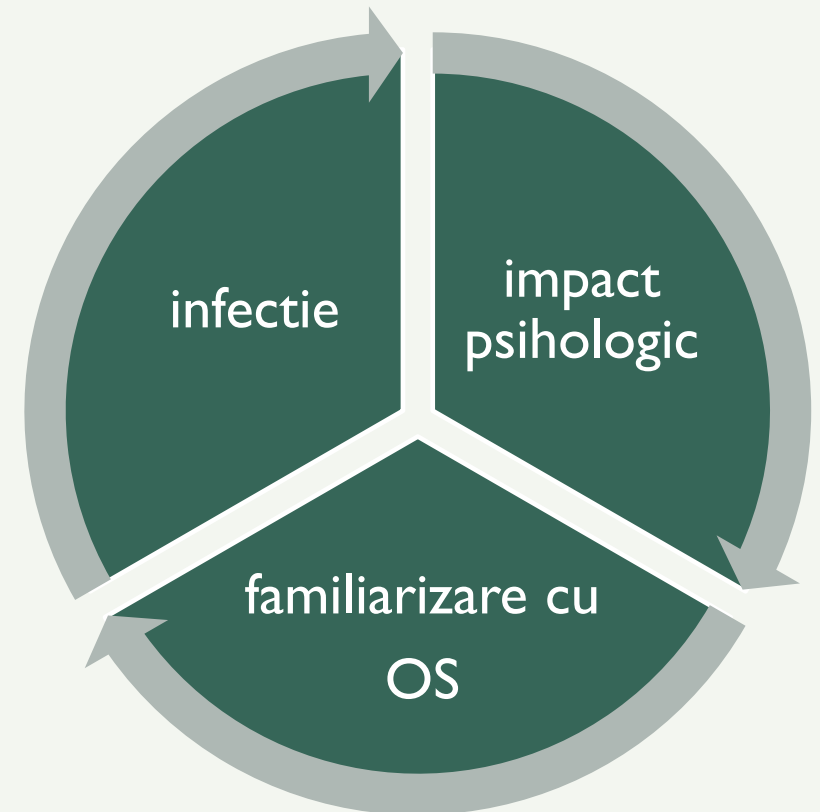


PERSISTENȚA SISTEMULUI DE OPERARE WINDOWS

RESPECTAREA LEGILOR NATURII

În mașinile biologice, virus+host=evoluție

- Sistemele de operare care nu pot fi infectate, dispar!
- Infecția este o metodă evolutivă de îmbunătățire a sistemului de operare.
- Infecția a implicat utilizatorul din punct de vedere psihologic!
- Utilizatorul se angajează și dedică timp sistemului de operare pentru a afla mai multe despre cum funcționează.
- Omul revine la sistemul de operare pe care îl cunoaște!

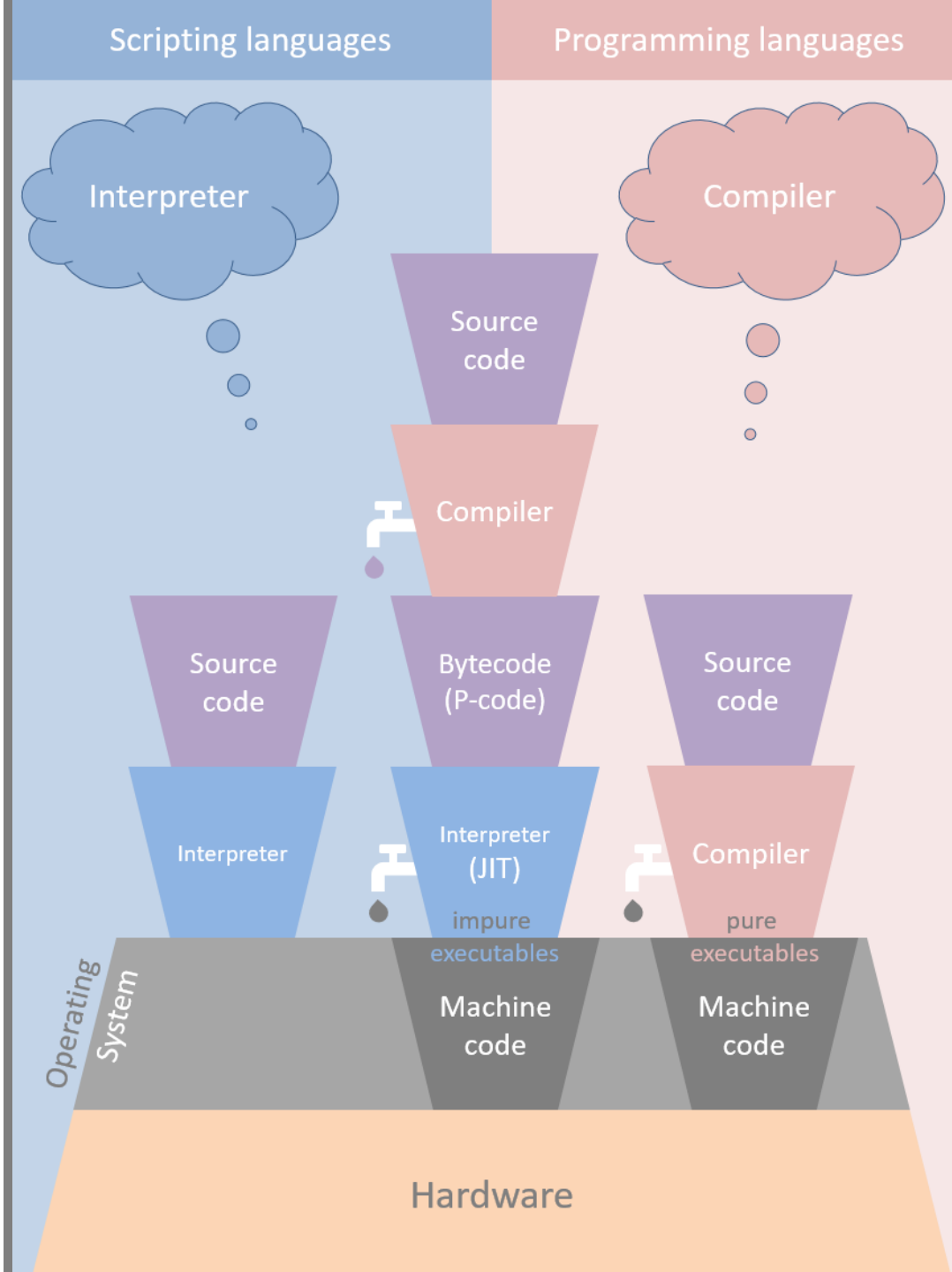


De ce ransomware este viitorul malware?

- Şurubul este strâns din punct de vedere al securităţii la nivel fundamental. Cum?:
- Sistem hybrid !

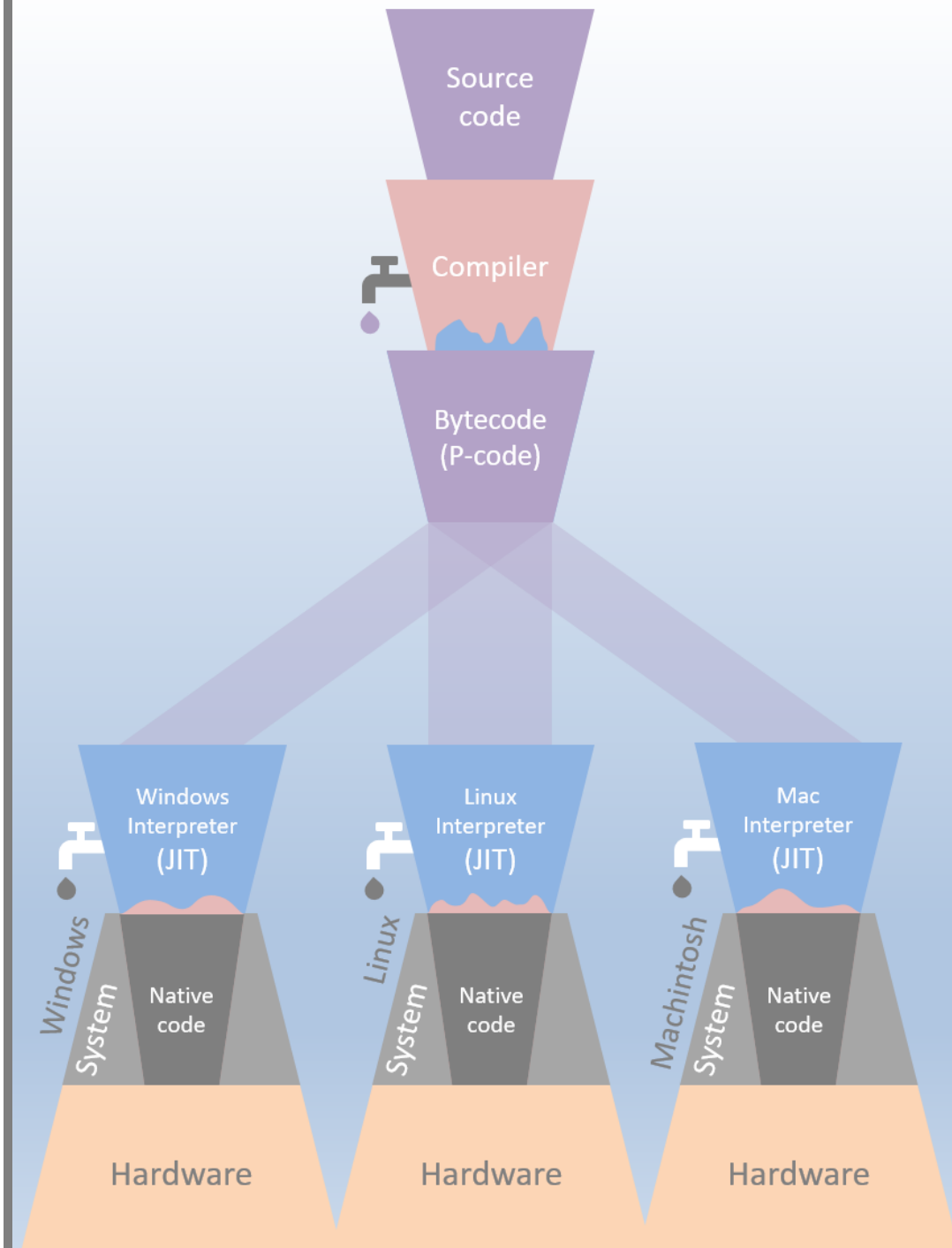
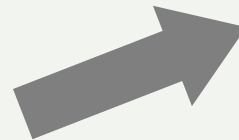
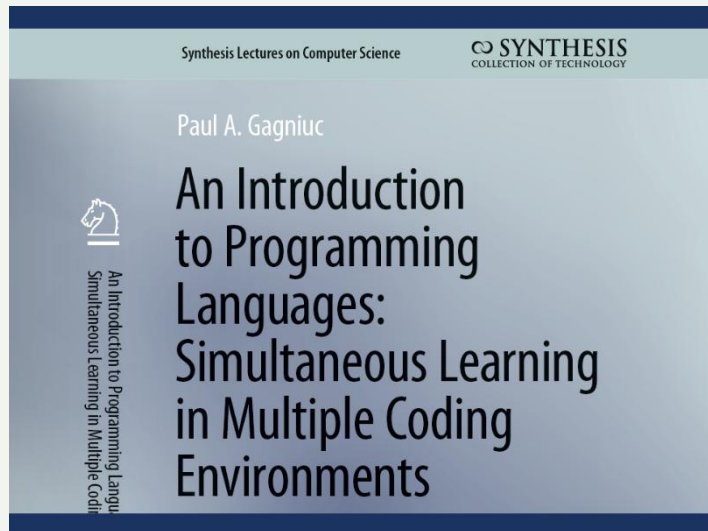
Se face distincţia între:

- Executabil – self (nativ)
 - Executabil - non-self (hibrid)
-
- Dezavantajul:
 - Sistemul hibrid rămâne mediocru.
 - Malware devine multi OS dar va fi non-self.



Direcția organizării software

- Universalitate:
- Limbaje de programare moderne
- Execuție pe mașini virtuale
- Portabilitatea aplicației



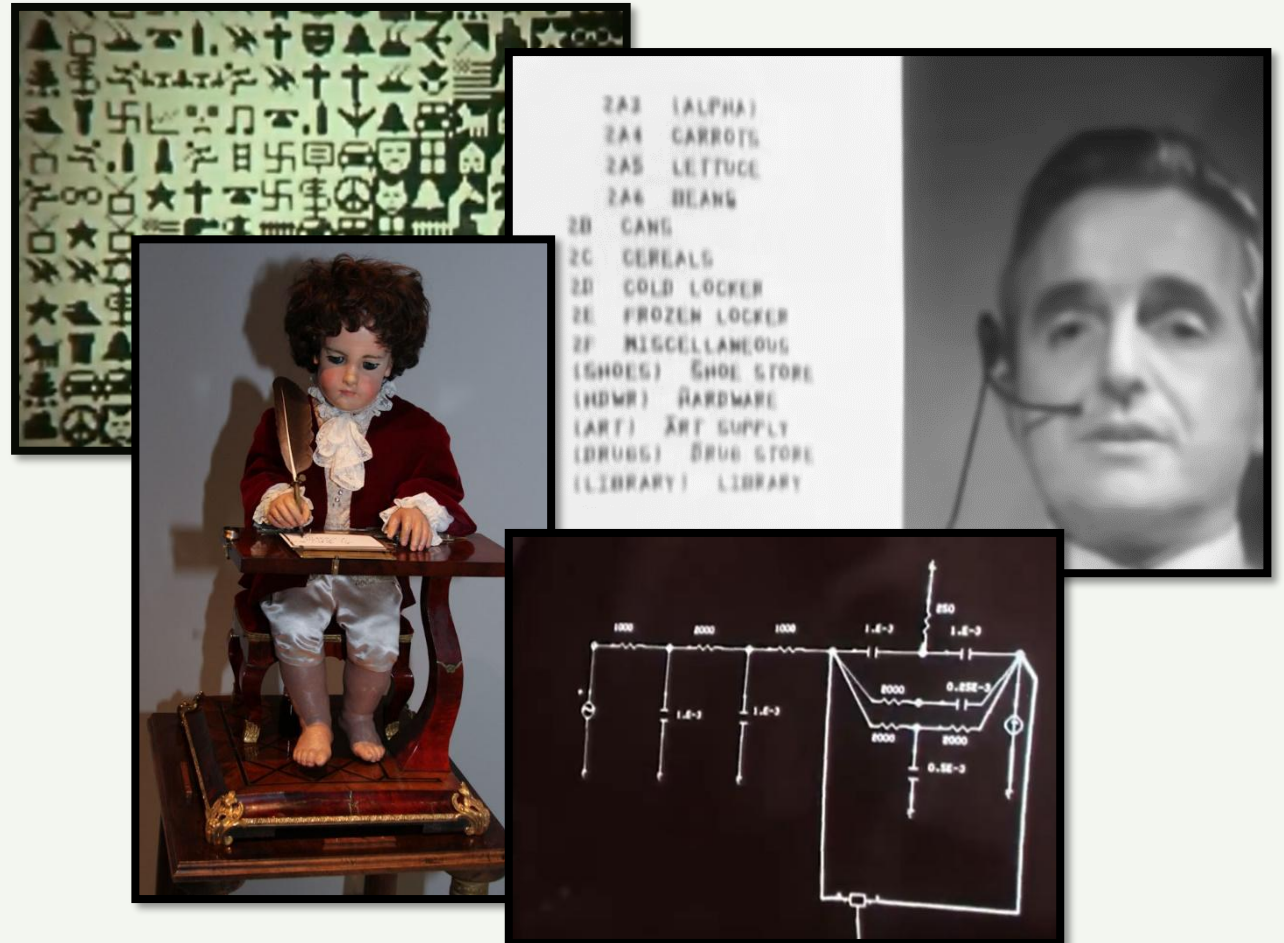
CONCLUZII

- Istoria calculatoarelor.
- Istoricul malware-ului și al soluțiilor de securitate.
- Evoluția tipurilor de malware.
- Evoluția sistemelor de operare, în special al Windows OS.
- Impactul psihologic al aplicațiilor malware asupra populației.
- Impactul sistemelor de operare asupra tipurilor de malware.

LINK-URI VIDEO RELEVANTE



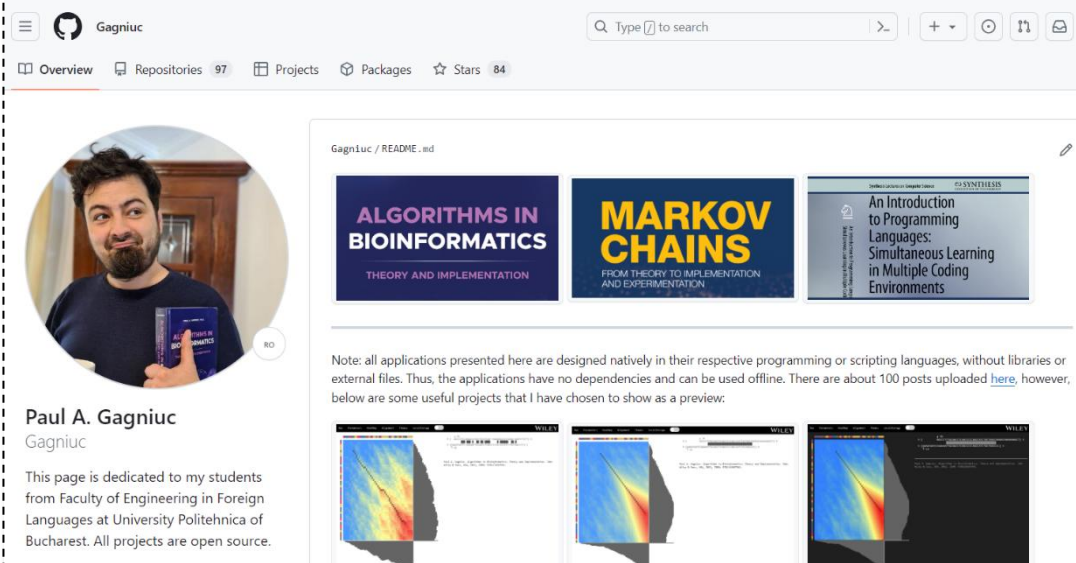
- [The Writer Automaton \(1774\)](#)
- [Mother of All Demos - 1968 \(1/3\)](#)
- [Mother of All Demos - 1968 \(2/3\)](#)
- [Mother of All Demos - 1968 \(3/3\)](#)
- [The Incredible Machine \(1968\)](#)



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- Paul A. Gagniuc. *An Introduction to Programming Languages: Simultaneous Learning in Multiple Coding Environments. Synthesis Lectures on Computer Science*. Springer International Publishing, 2023, pp. 1-280.
- Paul A. Gagniuc. *Coding Examples from Simple to Complex - Applications in MATLAB*, Springer, 2024, pp. 1-255.
- Paul A. Gagniuc. *Coding Examples from Simple to Complex - Applications in Python*, Springer, 2024, pp. 1-245.
- Paul A. Gagniuc. *Coding Examples from Simple to Complex - Applications in Javascript*, Springer, 2024, pp. 1-240.
- Paul A. Gagniuc. *Markov chains: from theory to implementation and experimentation*. Hoboken, NJ, John Wiley & Sons, USA, 2017, ISBN: 978-1-119-38755-8.

<https://github.com/gagniuc>



The screenshot shows the GitHub profile of Gagniuc. The header includes the GitHub logo, the name 'Gagniuc', a search bar, and navigation links for Overview, Repositories (97), Projects, Packages, Stars (84), and a mail icon. The profile picture shows a man with a beard holding a book. Below the profile picture, the name 'Paul A. Gagniuc' and the handle '@gagniuc' are displayed. A bio states: 'This page is dedicated to my students from Faculty of Engineering in Foreign Languages at University Politehnica of Bucharest. All projects are open source.' The main content area shows a 'README.md' file with three book covers: 'ALGORITHMS IN BIOINFORMATICS', 'MARKOV CHAINS', and 'An Introduction to Programming Languages: Simultaneous Learning in Multiple Coding Environments'. A note below the covers states: 'Note: all applications presented here are designed natively in their respective programming or scripting languages, without libraries or external files. Thus, the applications have no dependencies and can be used offline. There are about 100 posts uploaded here, however, below are some useful projects that I have chosen to show as a preview.' At the bottom, three small images of heatmaps or data visualizations are shown.